

AXI4-Compliant Memory Slave

UVM Verification Architecture

Comprehensive verification suite featuring dual-agent topology, SVA assertion invariants, functional coverage closures, and runtime factory component overrides.

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CHAPTER 1

Verification Strategy & Architecture

1.1 System Specifications and Constraints

The verified subsystem comprises an AMBA AXI4 Slave Bridge (`axi4.v`) attached to a Memory (`axi4_memory.v`). The operational parameters defined in `axi_para_pkg` and `parameters_pkg` govern configuration:

- **Data Bus Width (`DATA_WIDTH`):** 32 bits (4 bytes per word transfer).
- **Address Width (`ADDR_WIDTH`):** 16 bits externally; mapped internally to 10 bits ($2^{10} = 1024$ depth).
- **4KB Boundary Protection:** Transactions cannot cross 4KB address boundaries without returning `SLVERR` (`2'b10`).
- **Read Latency Pipeline:** Synchronous read latency accommodated via intermediate controller wait sequencing (`R_WAIT`).

1.2 Dual-Agent Topology

Figure 1.1 outlines the complete UVM environment topology. The testbench segregates protocol-level transactions from storage-level transactions using an active/passive agent split:

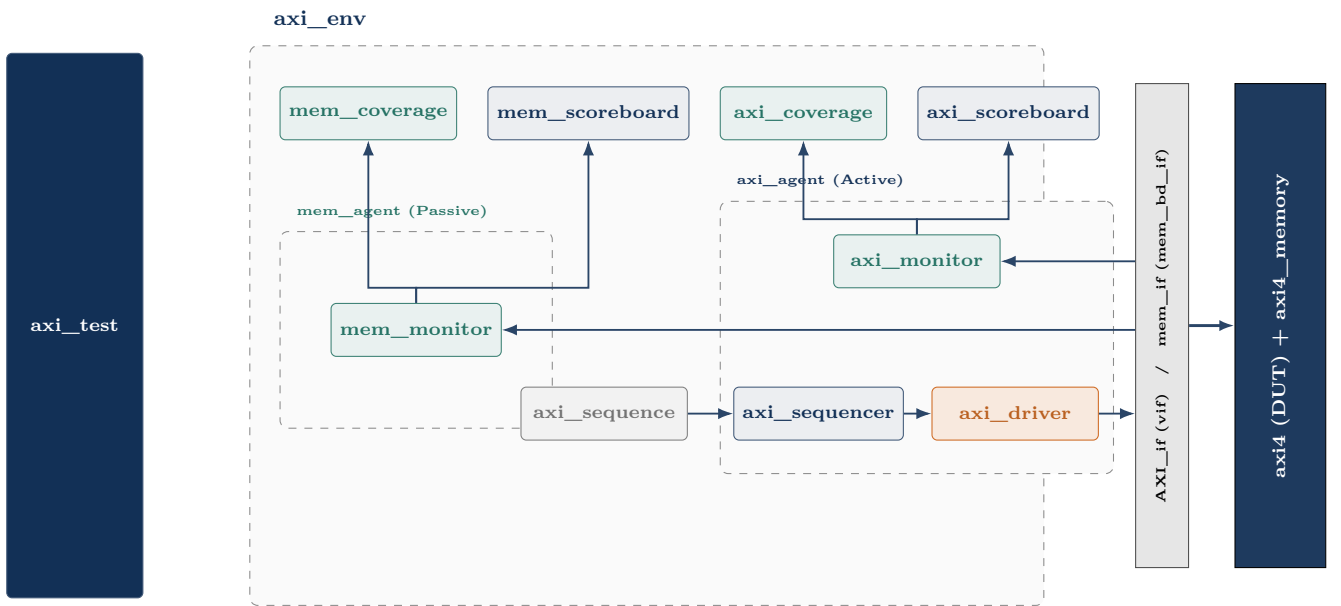


Figure 1.1: Dual-agent UVM environment architecture showing active AXI and passive memory paths.

- **Active AXI Agent (`axi_agent`):** Owns the sequencer (`axi_sequencer`), pin-level driver (`axi_driver` / `axi_driver_delay`), and AXI bus monitor (`axi_monitor`). It samples AW, W, B, AR, and R channels, converting transitions into completed transactions broadcast over `ap`.
- **Passive Memory Agent (`mem_agent`):** Contains exclusively `mem_monitor`. It attaches via interface bind (`mem_bd_if`) to observe raw memory enable, write-enable, address, and write/read data buses.

CHAPTER 2

Item Definition & Sequence Generation

2.1 Interfaces & Parameters

```
1  import axi_para_pkg::*;
2  import AXI_properties_pkg::*;
3
4  interface AXI_if #(
5      parameter DATA_WIDTH = 32,
6      parameter ADDR_WIDTH = 16,
7      parameter MEMORY_DEPTH = 1024
8  );
9
10     op_t                                operation;
11     logic                                ARESETn;
12     logic                                ACLK;
13     // Write address channel
14     logic [ADDR_WIDTH-1:0]              AWADDR;
15     logic [7:0]                          AWLEN;
16     logic [2:0]                          AWSIZE;
17     logic                                AWVALID;
18     logic                                AWREADY;
19
20     // Write data channel
21     logic [DATA_WIDTH-1:0]              WDATA;
22     logic                                WVALID;
23     logic                                WLAST;
24     logic                                WREADY;
25
26     // Write response channel
27     logic [1:0]                          BRESP;
28     logic                                BVALID;
29     logic                                BREADY;
30
31     // Read address channel
32     logic [ADDR_WIDTH-1:0]              ARADDR;
33     logic [7:0]                          ARLEN;
34     logic [2:0]                          ARSIZE;
35     logic                                ARVALID;
36     logic                                ARREADY;
37
38     // Read data channel
39     logic [DATA_WIDTH-1:0]              RDATA;
40     logic [1:0]                          RRESP;
41     logic                                RVALID;
42     logic                                RLAST;
43     logic                                RREADY;
44
45
46
47
48
49     // --- ASSERTIONS
50
51     Assertion_bvalid_after_last: assert property (
52         p_bvalid_after_wlast(ACLK, ARESETn, BVALID, WVALID, WREADY, WLAST)
53     ) else $error("[SVA ERROR] bvalid_after_last is failed");
54
55     Assertion_rready_after_last: assert property (
56         RREADY_after_wlast_is_zero(ACLK, ARESETn, RLAST, RREADY, RVALID)
57     ) else $error("[SVA ERROR] RREADY_after_wlast_is_zero is failed");
58
59     Assertion_awready_after_awvalid: assert property (
60         awready_after_awvalid(ACLK, ARESETn, AWVALID, AWREADY)
61     ) else $error("[SVA ERROR] awready_after_awvalid is failed");
62
63     Assertion_WLAST_AND_RLAST: assert property (
64         WLAST_AND_RLAST(ACLK, ARESETn, WLAST, RLAST)
65     ) else $error("[SVA ERROR] WLAST_AND_RLAST is failed");
66
67     Assertion_WLAST_AND_AWREADY: assert property (
68         WLAST_AND_AWREADY(ACLK, ARESETn, WLAST, AWREADY)
69     ) else $error("[SVA ERROR] WLAST_AND_AWREADY is failed");
70
71     Assertion_RLAST_AND_ARREADY: assert property (
72         RLAST_AND_ARREADY(ACLK, ARESETn, RLAST, ARREADY)
73     ) else $error("[SVA ERROR] RLAST_AND_ARREADY is failed");
74
75     Assertion_WLAST_AND_AWVALID: assert property (
76         WLAST_AND_AWVALID(ACLK, ARESETn, WLAST, AWVALID)
77     ) else $error("[SVA ERROR] WLAST_AND_AWVALID is failed");
```

```

78
79     Assertion_RLAST_AND_ARVALID: assert property (
80         RLAST_AND_ARVALID(ACLK, ARESETn, RLAST, ARVALID)
81     ) else $error("[SVA ERROR] RLAST_AND_ARVALID is failed");
82
83
84
85
86 endinterface

```

Listing 2.1: AXI Interface

```

1 package axi_para_pkg;
2     parameter DATA_WIDTH = 32;
3     parameter ADDR_WIDTH = 16;
4     parameter MEMORY_DEPTH = 1024;
5
6     typedef enum
7     {
8         WRITE,
9         READ,
10        IDLE
11    } op_t;
12 endpackage

```

Listing 2.2: AXI Parameters

```

1 package parameters_pkg;
2     parameter DATA_WIDTH = 32;
3     parameter ADDR_WIDTH = 10; // For 1024 locations
4     parameter DEPTH = 1024;
5
6 endpackage

```

Listing 2.3: Memory Parameters

```

1 package AXI_properties_pkg;
2
3     property p_bvalid_after_wlast(ACLK, ARESETn, BVALID, WVALID, WREADY, WLAST);
4     @(posedge ACLK) disable iff (!ARESETn)
5     $rose(BVALID) |-> $past(WVALID && WREADY && WLAST);
6     endproperty
7
8     property RREADY_after_wlast_is_zero(ACLK, ARESETn, RLAST, RREADY, RVALID);
9     @(posedge ACLK) disable iff (!ARESETn)
10    (RLAST && RREADY && RVALID) |-> #1 (!RREADY);
11    endproperty
12
13    property awready_after_awvalid(ACLK, ARESETn, AWVALID, AWREADY);
14    @(posedge ACLK) disable iff (!ARESETn)
15    AWVALID |-> #1 (!AWREADY);
16    endproperty
17
18    property WLAST_AND_RLAST(ACLK, ARESETn, WLAST, RLAST);
19    @(posedge ACLK) disable iff (!ARESETn)
20    not (WLAST && RLAST);
21    endproperty
22
23    property WLAST_AND_AWREADY(ACLK, ARESETn, WLAST, AWREADY);
24    @(posedge ACLK) disable iff (!ARESETn)
25    not (WLAST && AWREADY);
26    endproperty
27
28    property RLAST_AND_ARREADY(ACLK, ARESETn, RLAST, ARREADY);
29    @(posedge ACLK) disable iff (!ARESETn)
30    not (RLAST && ARREADY);
31    endproperty
32
33    property WLAST_AND_AWVALID(ACLK, ARESETn, WLAST, AWVALID);
34    @(posedge ACLK) disable iff (!ARESETn)
35    not (WLAST && AWVALID);
36    endproperty
37
38    property RLAST_AND_ARVALID(ACLK, ARESETn, RLAST, ARVALID);
39    @(posedge ACLK) disable iff (!ARESETn)
40    not (RLAST && ARVALID);
41    endproperty
42
43
44
45 endpackage

```

Listing 2.4: AXI Properties

```

1 package common_cfg_pkg;
2
3 import uvm_pkg::*;
4 `include "uvm_macros.svh"
5
6 class common_cfg extends uvm_object ;
7
8     'uvm_object_utils (common_cfg)
9     event stimulus_sent_e;
10
11     function new (string name = "common_cfg");
12         super.new(name);
13         'uvm_info(get_type_name(),"INSIDE NEW Common CLASS",UVM_MEDIUM)
14     endfunction
15
16 endclass
17
18 endpackage

```

Listing 2.5: Common Cfg

2.2 AXI Transaction Item (axi_transaction)

The primary transaction item, `axi_transaction`, defines both stimulus fields and response fields for single-beat and burst transfers. Dynamic arrays (`WDATA[]`, `RDATA[]`) capture full payload depth based on `AWLEN` and `ARLEN`.

```

1 package axi_transaction_pkg;
2
3 import axi_para_pkg::*;
4 import uvm_pkg::*;
5 `include "uvm_macros.svh"
6
7 class axi_transaction extends uvm_sequence_item;
8
9     rand logic ARESETn;
10    rand logic [ADDR_WIDTH-1:0] AWADDR;
11    rand logic [7:0] AWLEN;
12    logic [2:0] AWSIZE;
13    rand logic AWVALID;
14    logic AWREADY;
15    // Write data channel
16    rand logic [DATA_WIDTH-1:0] WDATA[];
17    logic WVALID;
18    logic WLAST;
19    logic WREADY;
20
21    // Write response channel
22    logic [1:0] BRESP;
23    logic BVALID;
24    logic BREADY;
25
26    // Read address channel
27    rand logic [ADDR_WIDTH-1:0] ARADDR;
28    rand logic [7:0] ARLEN;
29    logic [2:0] ARSIZE;
30    rand logic ARVALID;
31    logic ARREADY;
32
33    // Read data channel
34    logic [DATA_WIDTH-1:0] RDATA[];
35    logic RRESP;
36    logic RVALID;
37    logic RLAST;
38    logic RREADY;
39    rand logic [1:0] addr_mode;
40    rand logic len_mode;
41    rand op_t op;
42
43    constraint operation {
44        op dist {
45            WRITE := 35,
46            READ := 35,
47            IDLE := 30
48        };
49    }
50    constraint rst
51    {
52        ARESETn dist {1:=95 , 0:=5};
53    }
54
55    constraint address_mode
56    {
57        addr_mode dist {3:=10, 2:=20, 1:=40, 0:=30}; //3:=10, 2:=20, 1:=40, 0:=30
58    }
59    constraint length_mode

```

```

60     {
61     len_mode dist { 1:=50, 0:=50};
62     }
63
64     constraint addr {
65     if(addr_mode == 0)
66     {
67         AWADDR inside {[3211 : 4095]};
68         ARADDR == AWADDR;
69     }
70     else if (addr_mode == 2'd1)
71     {
72         (AWADDR + ((AWLEN + 1) * 4)) <= 4096;
73         ARADDR == AWADDR;
74     }
75     else if (addr_mode == 2'd2)
76     {
77         AWADDR inside {[4096 : 65536]};
78         ARADDR == AWADDR;
79     }
80     else
81     {
82         AWADDR inside {2024}; //0 , 4092
83         ARADDR == AWADDR;
84     }
85     }
86
87     constraint burst_c {
88     if(!len_mode)
89     {
90         AWLEN inside {[0:$]};
91     }
92     else
93     {
94         AWLEN inside {0};
95     }
96     WDATA.size() == AWLEN + 1;
97     ARLEN == AWLEN;
98     }
99
100    constraint valid {
101    if (op == WRITE) {
102        AWVALID == 1;
103        ARVALID == 0;
104    } else
105    if (op == READ) {
106        AWVALID == 0;
107        ARVALID != AWVALID;
108    } else {
109        AWVALID == 0;
110        ARVALID == 0;
111    }
112    }
113
114
115    'uvm_object_utils_begin (axi_transaction)
116        // Reset
117        'uvm_field_int(ARESETn, UVM_DEFAULT)
118
119        // Write address channel
120        'uvm_field_int(AWADDR, UVM_DEFAULT)
121        'uvm_field_int(AWLEN, UVM_DEFAULT)
122        'uvm_field_int(AWSIZE, UVM_DEFAULT)
123        'uvm_field_int(AWVALID, UVM_DEFAULT)
124        'uvm_field_int(AWREADY, UVM_DEFAULT)
125
126        // Write data channel
127        'uvm_field_array_int(WDATA, UVM_DEFAULT)
128        'uvm_field_int(WVALID, UVM_DEFAULT)
129        'uvm_field_int(WLAST, UVM_DEFAULT)
130        'uvm_field_int(WREADY, UVM_DEFAULT)
131
132        // Write response channel
133        'uvm_field_int(BRESP, UVM_DEFAULT)
134        'uvm_field_int(BVALID, UVM_DEFAULT)
135        'uvm_field_int(BREADY, UVM_DEFAULT)
136
137        // Read address channel
138        'uvm_field_int(ARADDR, UVM_DEFAULT)
139        'uvm_field_int(ARLEN, UVM_DEFAULT)
140        'uvm_field_int(ARSIZE, UVM_DEFAULT)
141        'uvm_field_int(ARVALID, UVM_DEFAULT)
142        'uvm_field_int(ARREADY, UVM_DEFAULT)
143
144        // Read data channel
145        'uvm_field_array_int(RDATA, UVM_DEFAULT)
146        'uvm_field_int(RRESP, UVM_DEFAULT)
147        'uvm_field_int(RVALID, UVM_DEFAULT)
148        'uvm_field_int(RLAST, UVM_DEFAULT)

```

```

149         'uvm_field_int(RREADY,          UVM_DEFAULT)
150
151         'uvm_object_utils_end
152
153         function new (string name = "axi_transaction");
154             super.new(name);
155         endfunction
156
157     endclass
158
159 endpackage

```

Listing 2.6: a_transaction.sv — AXI transaction definition with distribution constraints.

2.3 Memory Transaction Item (mem_transaction)

The passive memory transaction captures cycle-level signals at the boundary of the internal SRAM module.

```

1  package mem_transaction_pkg;
2
3      import uvm_pkg::*;
4      'include "uvm_macros.svh"
5      parameter MEM_ADDR_WIDTH = 10;
6      parameter MEM_DATA_WIDTH = 32;
7
8      class mem_transaction extends uvm_sequence_item;
9
10         logic                rst_n;
11         logic                mem_en;
12         logic                mem_we;
13         logic [MEM_ADDR_WIDTH-1:0] mem_addr;
14         logic [MEM_DATA_WIDTH-1:0] mem_wdata;
15         logic [MEM_DATA_WIDTH-1:0] mem_rdata;
16         'uvm_object_utils_begin(mem_transaction)
17             'uvm_field_int(mem_en, UVM_DEFAULT)
18             'uvm_field_int(mem_we, UVM_DEFAULT)
19             'uvm_field_int(mem_addr, UVM_DEFAULT)
20             'uvm_field_int(mem_wdata, UVM_DEFAULT)
21             'uvm_field_int(mem_rdata, UVM_DEFAULT)
22         'uvm_object_utils_end
23
24         function new(string name = "mem_transaction");
25             super.new(name);
26         endfunction
27
28     endclass
29
30 endpackage

```

Listing 2.7: mem_transaction.sv — Memory side transaction item.

2.4 Stimulus Generation (axi_sequence)

Stimulus generation runs inside `axi_sequence`, repeating 2000 randomized iterations across write, read, and idle cycles.

```
1 package axi_sequence_pkg;
2
3 import uvm_pkg::*;
4 import axi_transaction_pkg::*;
5 `include "uvm_macros.svh"
6
7 class axi_sequence extends uvm_sequence #(axi_transaction);
8
9     `uvm_object_utils(axi_sequence)
10
11     function new (string name = "axi_sequence");
12         super.new(name);
13         `uvm_info(get_type_name(), "AXI SEQUENCE INSIDE NEW", UVM_MEDIUM)
14     endfunction
15
16     task body();
17         axi_transaction req;
18         repeat(2000) begin
19             req = axi_transaction::type_id::create("req");
20             start_item(req);
21             assert(req.randomize());
22             `uvm_info(get_type_name(), {"DATA RANDOMIZED:"}, UVM_MEDIUM)
23             finish_item(req);
24
25         end
26     endtask
27
28 endclass
29
30 endpackage
```

Listing 2.8: `a_sequence.sv` — Sequence generating 2000 random transfers.

CHAPTER 3

Active AXI Agent Implementation

3.1 Driver Operations & Event Synchronization

The driver implements physical protocol handshakes on negative clock edges to maximize setup margins for the DUT. It uses an event handle (`stimulus_sent_e`) inside `common_cfg` to signal valid address assertion to the monitor.

```
1 package axi_driver_pkg;
2
3 import uvm_pkg::*;
4 import axi_para_pkg::*;
5 import axi_transaction_pkg::*;
6 import common_cfg_pkg::*;
7 'include "uvm_macros.svh"
8
9 class axi_driver extends uvm_driver #(axi_transaction);
10
11     shortint i = 0;
12     shortint j = 0;
13     virtual AXI_if axi_vif;
14     common_cfg m_cfg;
15     'uvm_component_utils(axi_driver)
16
17     function new (string name = "axi_driver" , uvm_component parent);
18         super.new(name, parent);
19     endfunction
20
21     function void build_phase (uvm_phase phase);
22         super.build_phase(phase);
23         if(!(uvm_config_db #(virtual AXI_if)::get(this, "", "vif", axi_vif)))
24             'uvm_fatal(get_type_name(), "FAILED TO GET INTERFERENCE")
25         'uvm_info(get_type_name(), "AXI DRIVER BUILD PHASE", UVM_MEDIUM)
26     endfunction
27
28     task run_phase(uvm_phase phase);
29     forever begin
30         axi_transaction req;
31         'uvm_info("DRV", "Waiting for transaction", UVM_MEDIUM)
32         seq_item_port.get_next_item(req);
33         'uvm_info("DRV", "Got transaction", UVM_MEDIUM)
34         drive(req);
35         'uvm_info("DRV", "Drive finished", UVM_MEDIUM)
36         seq_item_port.item_done();
37     end
38 endtask
39
40     extern task drive (axi_transaction req);
41
42 endclass
43
44 task axi_driver::drive (axi_transaction req);
45     @(negedge axi_vif.ACLK);
46     req.print();
47
48     axi_vif.ARESETn <= req.ARESETn;
49     if(!req.ARESETn)
50     begin
51         'uvm_info(get_type_name(), ">>> DRIVER ASSERTING RESET <<<", UVM_LOW)
52         // Reset all outputs
53         axi_vif.AWREADY <= 1'b1; // Ready to accept address
54         axi_vif.WREADY <= 1'b0;
55         axi_vif.BVALID <= 1'b0;
56         axi_vif.BRESP <= 2'b00;
57
58         axi_vif.ARREADY <= 1'b1; // Ready to accept address
59         axi_vif.RVALID <= 1'b0;
60         axi_vif.RRESP <= 2'b00;
61         axi_vif.RDATA <= {DATA_WIDTH{1'b0}};
62         axi_vif.RLAST <= 1'b0;
63         @(negedge axi_vif.ACLK);
64         -> m_cfg.stimulus_sent_e;
65
66     end
67     else
68     begin
69         case(req.op)
70             WRITE:
71                 begin
72                     @(negedge axi_vif.ACLK);
73                     axi_vif.AWVALID <= req.AWVALID;
```

```

74         axi_vif.AWADDR <= req.AWADDR;
75         axi_vif.AWLEN  <= req.AWLEN;
76         axi_vif.AWSIZE <= 'd2;
77
78         'uvm_info(get_type_name(), "DRV Triggered", UVM_MEDIUM)
79         while (!axi_vif.AWREADY)
80             @(posedge axi_vif.ACLK);
81
82         @(negedge axi_vif.ACLK);
83         axi_vif.AWVALID <= 0;
84         -> m_cfg.stimulus_sent_e;
85
86         $display("WDATA size = %0d", req.WDATA.size());
87
88         foreach (req.WDATA[i]) begin
89
90             @(negedge axi_vif.ACLK);
91
92             axi_vif.WVALID <= 1;
93             axi_vif.BREADY <= 1;
94             axi_vif.WLAST  <= (i == req.WDATA.size()-1);
95             axi_vif.WDATA  <= req.WDATA[i];
96
97             while (!axi_vif.WREADY)
98                 @(posedge axi_vif.ACLK);
99
100             @(negedge axi_vif.ACLK);
101             axi_vif.WVALID <= 0;
102             axi_vif.WLAST  <= 0;
103
104         end
105
106         while (!axi_vif.BVALID)
107             @(posedge axi_vif.ACLK);
108         req.BRESP = axi_vif.BRESP;
109
110         @(negedge axi_vif.ACLK);
111         // #50ps;
112         axi_vif.BREADY <= 0;
113     end
114 READ:
115     begin
116         @(negedge axi_vif.ACLK);
117         //repeat(2) @(posedge axi_vif.ACLK);
118         axi_vif.ARADDR <= req.ARADDR;
119         axi_vif.ARLLEN <= req.ARLLEN;
120         axi_vif.ARVALID <= req.ARVALID;
121         axi_vif.ARSIZE <= 'd2;
122         axi_vif.AWVALID <= req.AWVALID;
123         axi_vif.AWADDR <= req.AWADDR;
124         axi_vif.AWLEN  <= req.AWLEN;
125         axi_vif.AWSIZE <= 'd2;
126
127         while (!axi_vif.ARREADY)
128             @(posedge axi_vif.ACLK);
129
130         @(negedge axi_vif.ACLK);
131         axi_vif.ARVALID <= 0;
132         -> m_cfg.stimulus_sent_e;
133
134         req.RDATA = new[req.ARLLEN+1];
135
136         foreach (req.RDATA[i])
137             begin
138
139                 @(negedge axi_vif.ACLK);
140                 axi_vif.RREADY <= 1;
141
142                 while (!axi_vif.RVALID)
143                     @(negedge axi_vif.ACLK);
144
145                 req.RDATA[i] = axi_vif.RDATA;
146                 req.RRESP    = axi_vif.RRESP;           // capture response while you're here
147
148             end
149
150             @(negedge axi_vif.ACLK);
151             axi_vif.RREADY <= 0;
152         end
153 IDLE:
154     begin
155         @(negedge axi_vif.ACLK);
156
157         axi_vif.AWVALID <= req.AWVALID;
158         axi_vif.WVALID  <= 0;
159         axi_vif.BREADY  <= 0;
160         axi_vif.AWADDR  <= req.AWADDR;
161         axi_vif.AWLEN   <= req.AWLEN;

```

```

163         axi_vif.AWSIZE  <= 'd2;
164
165         axi_vif.ARVALID <= req.ARVALID;
166         axi_vif.RREADY  <= 0;
167         axi_vif.WLAST   <= 0;
168         axi_vif.ARADDR  <= req.ARADDR;
169         axi_vif.ARLLEN  <= req.ARLLEN;
170         axi_vif.ARSIZE  <= 'd2;
171         -> m_cfg.stimulus_sent_e;
172
173         @(posedge axi_vif.ACLK);
174     end
175 endcase
176 end
177
178
179 endtask
180 endpackage

```

Listing 3.1: a_driver.sv — Master AXI bus pin-level driver.

3.2 Factory Override Driver (axi_driver_delay)

The derived class `axi_driver_delay` extends `axi_driver` to insert a randomized delay between 0 and 3 clock cycles ahead of calling `drive(req)`.

```

1  package axi_driver_delay_pkg;
2
3      import uvm_pkg::*;
4      import axi_para_pkg::*;
5      import axi_transaction_pkg::*;
6      import axi_driver_pkg::*;
7      import common_cfg_pkg::*;
8      'include "uvm_macros.svh"
9
10     class axi_driver_delay extends axi_driver;
11
12         'uvm_component_utils(axi_driver_delay)
13
14         function new(
15             string name = "axi_driver_delay",
16             uvm_component parent
17         );
18             super.new(name, parent);
19         endfunction
20
21         function void build_phase(uvm_phase phase);
22             super.build_phase(phase);
23
24             'uvm_info(
25                 get_type_name(),
26                 "AXI DELAY DRIVER BUILD PHASE",
27                 UVM_MEDIUM
28             )
29         endfunction
30
31         task run_phase(uvm_phase phase);
32             forever begin
33                 axi_transaction req;
34
35                 seq_item_port.get_next_item(req);
36
37                 // Random idle cycles
38                 repeat ($urandom_range(0,3))
39                     @(posedge axi_vif.ACLK);
40
41                 // Use parent's drive()
42                 drive(req);
43
44                 seq_item_port.item_done();
45             end
46         endtask
47
48     endclass
49
50 endpackage

```

Listing 3.2: a_delay_driver.sv — Overridden driver injecting idle timing cycles.

3.3 Axi sequencer.

```

1 package axi_sequencer_pkg;
2
3 import uvm_pkg::*;
4 import axi_transaction_pkg::*;
5 `include "uvm_macros.svh"
6
7 class axi_sequencer extends uvm_sequencer #(axi_transaction);
8
9     `uvm_component_utils(axi_sequencer)
10
11     function new (string name = "axi_sequencer" , uvm_component parent);
12         super.new(name, parent);
13     endfunction
14
15     function void build_phase (uvm_phase phase);
16         super.build_phase(phase);
17         `uvm_info(get_type_name(), "AXI SEQUENCER BUILD PHASE", UVM_MEDIUM)
18     endfunction
19
20 endclass
21 endpackage

```

Listing 3.3: Axi sequencer.

3.4 AXI Bus Monitor (axi_monitor)

The monitor samples all channel signals upon triggering `stimulus_sent_e`, decodes the address region into coverage bins, and broadcasts transactions through `ap`.

```

1 package axi_monitor_pkg;
2
3 import uvm_pkg::*;
4 import axi_para_pkg::*;
5 import common_cfg_pkg::*;
6 import axi_transaction_pkg::*;
7 `include "uvm_macros.svh"
8
9 class axi_monitor extends uvm_monitor;
10
11     uvm_analysis_port #(axi_transaction) ap;
12     virtual AXI_if axi_vif;
13     common_cfg m_cfg;
14     `uvm_component_utils(axi_monitor)
15
16     function new (string name = "axi_monitor" , uvm_component parent);
17         super.new(name, parent);
18         ap = new("ap", this);
19     endfunction
20
21     function void build_phase (uvm_phase phase);
22         super.build_phase(phase);
23         if (!(uvm_config_db #(virtual AXI_if)::get(this, "", "vif", axi_vif)))
24             `uvm_fatal(get_type_name(), "FAILED TO GET INTERFERENCE")
25         `uvm_info(get_type_name(), "AXI MONITOR BUILD PHASE", UVM_LOW)
26     endfunction
27
28     task run_phase(uvm_phase phase);
29         int i = 0;
30         `uvm_info(get_type_name(), "Entered Run phase in monitor", UVM_LOW)
31         forever
32             begin
33
34                 wait(m_cfg.stimulus_sent_e.triggered)
35                 if (!axi_vif.ARESETn)
36                     begin
37                         axi_transaction tr = axi_transaction::type_id::create("tr_rst");
38                         tr.ARESETn = axi_vif.ARESETn;
39                         tr.AWREADY = axi_vif.AWREADY;
40                         tr.WREADY = axi_vif.WREADY;
41                         tr.BVALID = axi_vif.BVALID;
42                         tr.BRESP = axi_vif.BRESP;
43                         tr.ARREADY = axi_vif.ARREADY;
44                         tr.RVALID = axi_vif.RVALID;
45                         tr.RRESP = axi_vif.RRESP;
46                         tr.RLAST = axi_vif.RLAST;
47                         tr.RDATA = new[1];
48                         tr.RDATA[0] = axi_vif.RDATA;
49                         tr.addr_mode = 0;
50
51                         `uvm_info(get_type_name(), "Reset detected on bus: sending to scoreboard", UVM_LOW)
52                         ap.write(tr);
53                         @(posedge axi_vif.ACLK);
54                     end
55             end
56     endtask
57 endclass
58 endpackage

```

```

55
56     end
57     else if (axi_vif.AWVALID)
58     begin
59         axi_transaction tr = axi_transaction ::type_id::create("tr");
60         'uvm_info(get_type_name(), "MON Triggered - WRITE", UVM_LOW)
61         tr.op = WRITE;
62         tr.AWADDR = axi_vif.AWADDR;
63         tr.AWLEN = axi_vif.AWLEN;
64         tr.AWSIZE = axi_vif.AWSIZE;
65         tr.AWVALID = axi_vif.AWVALID;
66         tr.AWREADY = axi_vif.AWREADY;
67         tr.WDATA = new[axi_vif.AWLEN+1];
68
69         // For Coverage
70         if (tr.AWADDR >= 3211 && tr.AWADDR <= 4095)
71             tr.addr_mode = 2'd0;
72         else if ((tr.AWADDR + ((tr.AWLEN + 1) * 4)) <= 4096)
73             tr.addr_mode = 2'd1;
74         else if (tr.AWADDR >= 4096)
75             tr.addr_mode = 2'd2;
76         else
77             tr.addr_mode = 2'd3;
78
79
80         for (int x = 0 ; x<=tr.AWLEN;x++)
81         begin
82             @(posedge axi_vif.ACLK);
83             @(posedge axi_vif.ACLK);
84             tr.WDATA[x] = axi_vif.WDATA;
85             tr.WVALID = axi_vif.WVALID;
86             tr.WLAST = axi_vif.WLAST;
87             tr.WREADY = axi_vif.WREADY;
88             tr.BRESP = axi_vif.BRESP;
89             tr.BVALID = axi_vif.BVALID;
90             tr.BREADY = axi_vif.BREADY;
91             'uvm_info("MON", "About to send transaction Write OP", UVM_LOW)
92             ap.write(tr);
93             'uvm_info("MON", "Transaction sent Write OP", UVM_LOW)
94         end
95         @(posedge axi_vif.ACLK);
96         wait(axi_vif.BVALID);
97         tr.BRESP = axi_vif.BRESP;
98         tr.BVALID = axi_vif.BVALID;
99         @(posedge axi_vif.ACLK);
100        tr.BREADY = axi_vif.BREADY;
101        'uvm_info("MON", "About to send transaction", UVM_LOW)
102        ap.write(tr);
103        'uvm_info("MON", "Transaction sent", UVM_LOW)
104    end
105    else if (axi_vif.ARVALID)
106    begin
107        axi_transaction tr = axi_transaction ::type_id::create("tr");
108        'uvm_info(get_type_name(), "MON Triggered - READ", UVM_LOW)
109        tr.op = READ;
110        tr.ARADDR = axi_vif.ARADDR;
111        tr.ARLLEN = axi_vif.ARLLEN;
112        tr.ARSIZE = axi_vif.ARSIZE;
113        tr.ARVALID = axi_vif.ARVALID;
114        tr.ARREADY = axi_vif.ARREADY;
115        tr.RDATA = new[axi_vif.ARLLEN+1];
116
117        // For Coverage
118        if (tr.ARADDR >= 3211 && tr.ARADDR <= 4095)
119            tr.addr_mode = 2'd0;
120        else if ((tr.ARADDR + ((tr.ARLLEN + 1) * 4)) <= 4096)
121            tr.addr_mode = 2'd1;
122        else if (tr.ARADDR >= 4096)
123            tr.addr_mode = 2'd2;
124        else
125            tr.addr_mode = 2'd3;
126
127        @(posedge axi_vif.ACLK);
128        @(posedge axi_vif.ACLK);
129        for (int x = 0 ; x<=tr.ARLLEN;x++)
130        begin
131            @(posedge axi_vif.ACLK);
132            @(posedge axi_vif.ACLK);
133            tr.RDATA[x] = axi_vif.RDATA;
134            tr.RVALID = axi_vif.RVALID;
135            tr.RLAST = axi_vif.RLAST;
136            tr.RREADY = axi_vif.RREADY;
137            tr.RRESP = axi_vif.RRESP;
138            'uvm_info("MON", "About to send transaction Read OP", UVM_LOW)
139            ap.write(tr);
140            'uvm_info("MON", "Transaction sent Read OP", UVM_LOW)
141        end
142    end
143    else

```

```

144         begin
145             axi_transaction tr = axi_transaction ::type_id::create("tr");
146             'uvm_info(get_type_name(), "MON Triggered - IDLE", UVM_LOW)
147             @(posedge axi_vif.ACLK);
148             tr.op = IDLE;
149             tr.ARADDR = axi_vif.ARADDR;
150             tr.ARLLEN = axi_vif.ARLLEN;
151             tr.ARSIZE = axi_vif.ARSIZE;
152             tr.ARVALID = axi_vif.ARVALID;
153             tr.ARREADY = axi_vif.ARREADY;
154             tr.RDATA = new[1];
155             tr.RVALID = axi_vif.RVALID;
156             tr.RLAST = axi_vif.RLAST;
157             tr.RREADY = axi_vif.RREADY;
158             tr.RRESP = axi_vif.RRESP;
159             tr.AWADDR = axi_vif.AWADDR;
160             tr.AWLEN = axi_vif.AWLEN;
161             tr.AWSIZE = axi_vif.AWSIZE;
162             tr.AWVALID = axi_vif.AWVALID;
163             tr.AWREADY = axi_vif.AWREADY;
164             tr.WDATA = new[1];
165             tr.WVALID = axi_vif.WVALID;
166             tr.WLAST = axi_vif.WLAST;
167             tr.WREADY = axi_vif.WREADY;
168             tr.BRESP = axi_vif.BRESP;
169             tr.BVALID = axi_vif.BVALID;
170             tr.BREADY = axi_vif.BREADY;
171
172             // For Coverage //////////////////////////////////////
173             if (axi_vif.AWADDR >= 3211 && axi_vif.AWADDR <= 4095)
174                 tr.addr_mode = 2'd0;
175             else if ((axi_vif.AWADDR + ((axi_vif.AWLEN + 1) * 4)) <= 4096)
176                 tr.addr_mode = 2'd1;
177             else if (axi_vif.AWADDR >= 4096)
178                 tr.addr_mode = 2'd2;
179             else
180                 tr.addr_mode = 2'd3;
181             //////////////////////////////////////
182             'uvm_info("MON", "About to send transaction IDLE OP", UVM_LOW)
183             ap.write(tr);
184             'uvm_info("MON", "Transaction sent IDLE OP", UVM_LOW)
185             @(posedge axi_vif.ACLK);
186         end
187     end
188 endtask
189
190
191
192 endclass
193 endpackage

```

Listing 3.4: a_monitor.sv — Channel sampling monitor.

CHAPTER 4

Agents Implementation

4.1 Direct Interface Binding

To verify internal memory interactions without intruding on RTL source files, `mem_if` binds directly to `axi4_memory` inside `top.sv`.

```
1 interface mem_if #(
2     parameter DATA_WIDTH = 32,
3     parameter ADDR_WIDTH = 10,
4     parameter DEPTH = 1024
5 )(
6     input logic          clk,
7     input logic          rst_n,
8     input logic          mem_en,
9     input logic          mem_we,
10    input logic [ADDR_WIDTH-1:0] mem_addr,
11    input logic [DATA_WIDTH-1:0] mem_wdata,
12    input logic [DATA_WIDTH-1:0] mem_rdata
13 );
14 endinterface
```

Listing 4.1: Memory interface.

```
1 bind axi4_memory mem_if #(
2     .ADDR_WIDTH(10),
3     .DATA_WIDTH(32)
4 ) mem_bd_if (
5     .clk      (clk),
6     .rst_n    (rst_n),
7     .mem_en   (mem_en),
8     .mem_we   (mem_we),
9     .mem_addr (mem_addr),
10    .mem_wdata(mem_wdata),
11    .mem_rdata(mem_rdata)
12 );
```

Listing 4.2: Top module binding showing passive interface attachment.

4.2 Memory Monitor (mem_monitor) & Agent (mem_agent)

The passive monitor samples internal access signals on the rising clock edge plus a 10 ps settling offset.

```

1  package mem_monitor_pkg;
2
3      import uvm_pkg::*;
4      import common_cfg_pkg::*;
5      import mem_transaction_pkg::*;
6      'include "uvm_macros.svh"
7
8      class mem_monitor extends uvm_monitor;
9
10         uvm_analysis_port #(mem_transaction) mem_ap;
11         virtual mem_if mem_vif;
12         common_cfg m_cfg;
13         'uvm_component_utils(mem_monitor)
14
15         function new (string name = "mem_monitor" , uvm_component parent);
16             super.new(name, parent);
17             mem_ap = new("mem_ap",this);
18         endfunction
19         function void build_phase (uvm_phase phase);
20             super.build_phase(phase);
21             if(! (uvm_config_db #(virtual mem_if)::get(this, "", "mem_vif", mem_vif)))
22                 'uvm_fatal(get_type_name(), "FAILED TO GET MEM INTERFERENCE")
23
24             'uvm_info(get_type_name(), "MEM MONITOR BUILD PHASE", UVM_LOW)
25         endfunction
26         task run_phase(uvm_phase phase);
27             int i = 0;
28             'uvm_info(get_type_name(), "Entered Run phase in MEM monitor", UVM_LOW)
29             forever
30                 begin
31                     mem_transaction mem_tr = mem_transaction::type_id::create("mem_tr");
32                     mem_tr.mem_en      = mem_vif.mem_en;
33                     mem_tr.mem_we      = mem_vif.mem_we;
34                     mem_tr.mem_addr     = mem_vif.mem_addr;
35                     mem_tr.mem_wdata   = mem_vif.mem_wdata;
36                     mem_tr.rst_n       = mem_vif.rst_n;
37                     @(posedge mem_vif.clk);
38                     #10ps;
39                     mem_tr.mem_rdata = mem_vif.mem_rdata;
40                     'uvm_info("MEM_MON", "About to send MEM transaction", UVM_LOW)
41                     mem_ap.write(mem_tr);
42                     'uvm_info("MEM_MON", "About to send MEM transaction ", UVM_LOW)
43                 end
44             endtask
45         endclass
46     endpackage

```

Listing 4.3: mem_monitor.sv implementation.

4.3 AXI & Memory Agent

```

1 package axi_agent_pkg;
2
3 //import axi_pkg::*;
4 import uvm_pkg::*;
5
6 import axi_driver_pkg::*;
7 import axi_driver_delay_pkg::*;
8 import axi_monitor_pkg::*;
9 import axi_sequencer_pkg::*;
10 'include "uvm_macros.svh"
11
12 class axi_agent extends uvm_agent;
13
14     axi_driver drv;
15     axi_monitor mon;
16     axi_sequencer sqr;
17
18     uvm_active_passive_enum is_active = UVM_ACTIVE;
19
20     'uvm_component_utils(axi_agent)
21
22     function new (string name = "axi_agent" , uvm_component parent);
23         super.new(name, parent);
24     endfunction
25
26     function void build_phase (uvm_phase phase);
27         super.build_phase(phase);
28         if(!(uvm_config_db #(uvm_active_passive_enum)::get(this, "", "is_active", is_active)))
29             'uvm_fatal(get_type_name(), "FAILED TO GET UVM ACTIVE PASSIVE AGENT")
30
31         mon = axi_monitor::type_id::create("mon", this);
32         if(is_active == UVM_ACTIVE)
33             begin
34                 drv = axi_driver::type_id::create("drv", this);
35                 sqr = axi_sequencer::type_id::create("sqr", this);
36             end
37         'uvm_info(get_type_name(), "AXI AGENT BUILD PHASE", UVM_MEDIUM)
38     endfunction
39
40     function void connect_phase (uvm_phase phase);
41         super.connect_phase(phase);
42         if(is_active == UVM_ACTIVE)
43             drv.seq_item_port.connect(sqr.seq_item_export);
44     endfunction
45
46 endclass
47 endpackage

```

Listing 4.4: a_agent.sv

```

1 package mem_agent_pkg;
2 import uvm_pkg::*;
3 import mem_monitor_pkg::*;
4 'include "uvm_macros.svh"
5
6 class mem_agent extends uvm_agent;
7
8     mem_monitor mem_mon;
9     'uvm_component_utils(mem_agent)
10
11     function new (string name = "mem_agent" , uvm_component parent);
12         super.new(name, parent);
13     endfunction
14
15     function void build_phase (uvm_phase phase);
16         super.build_phase(phase);
17         //if(!(uvm_config_db #(uvm_active_passive_enum)::get(this, "", "is_active", is_active)))
18             // 'uvm_fatal(get_type_name(), "FAILED TO GET UVM ACTIVE PASSIVE AGENT")
19
20         mem_mon = mem_monitor::type_id::create("mem_mon", this);
21         'uvm_info(get_type_name(), "MEM AGENT BUILD PHASE", UVM_MEDIUM)
22     endfunction
23 endclass
24 endpackage

```

Listing 4.5: mem_agent.sv implementations.

CHAPTER 5

Scoreboards & Functional Coverage

5.1 Dual Scoreboard Architecture

Data integrity checks operate on two levels concurrently:

- **AXI Bus Scoreboard (axi_scoreboard):** Tracks AXI protocol rules, validates BRESP/RRESP codes (2'b00 for OKAY, 2'b10 for SLVERR), detects 4KB boundaries, and flags invalid address bounds.
- **Memory Scoreboard (mem_scoreboard):** Maintains an independent shadow array (memory_golden[1024]) and checks read/write behavior directly against raw memory lines.

```
1 package axi_scoreboard_pkg;
2
3 import axi_transaction_pkg::*;
4 import axi_para_pkg::*;
5 import uvm_pkg::*;
6 `include "uvm_macros.svh"
7
8 class axi_scoreboard extends uvm_scoreboard;
9
10
11     uvm_analysis_export #(axi_transaction) analysis_export;
12     uvm_tlm_analysis_fifo #(axi_transaction) fifo;
13     int pass_total_count = 0;
14     int fail_total_count = 0;
15
16     `uvm_component_utils(axi_scoreboard)
17
18     reg [DATA_WIDTH-1:0] gmemory [0:MEMORY_DEPTH-1];
19
20     function new (string name = "axi_scoreboard" , uvm_component parent);
21         super.new(name, parent);
22         analysis_export = new("analysis_export",this);
23         fifo = new("fifo",this);
24     endfunction
25
26     function void build_phase (uvm_phase phase);
27         super.build_phase(phase);
28         `uvm_info(get_type_name(),"AXI SCOREBOARD BUILD PHASE",UVM_MEDIUM)
29
30         for (int z = 0; z < MEMORY_DEPTH; z = z + 1)
31             gmemory[z] = 0;
32
33     endfunction
34
35     function void connect_phase(uvm_phase phase);
36         analysis_export.connect(fifo.analysis_export);
37     endfunction
38
39
40     task run_phase(uvm_phase phase);
41         axi_transaction tr;
42         int i = 0;
43         int j = 0;
44         logic safeW;
45         logic safeR;
46         forever
47             begin
48
49                 `uvm_info(get_type_name(), "Waiting for transaction", UVM_LOW)
50                 fifo.get(tr);
51                 safeW = !(((tr.AWADDR & 12'hFFF) + ((tr.AWLEN + 1) << tr.AWSIZE)) > 12'hFFF) && ((tr.
52                     AWADDR + i * (1 << tr.AWSIZE)) >> 2) < MEMORY_DEPTH;
53                 safeR = !(((tr.ARADDR & 12'hFFF) + ((tr.ARLen + 1) << tr.ARSIZE)) > 12'hFFF) && ((tr.
54                     ARADDR + i * (1 << tr.ARSIZE)) >> 2) < MEMORY_DEPTH;
55                 `uvm_info(get_type_name(), "Transaction received", UVM_LOW)
56                 tr.print();
57                 if (!tr.ARESETn)
58                     begin
59                         i = 0;
60                         j = 0;
61
62                         // Check AWREADY
63                         if (tr.AWREADY === 1'b1) begin
64                             `uvm_info(get_type_name(), $sformatf("RESET CHECK | AWREADY Expected=1'b1 Actual
65                                 =%0b | PASS", tr.AWREADY), UVM_LOW)
66                             pass_total_count++;
67                         end else begin
68                             `uvm_error(get_type_name(), $sformatf("RESET CHECK | AWREADY Expected=1'b1 Actual
69                                 =%0b", tr.AWREADY))
70                         end
71                     end
72             end
73     endtask
74 endclass
```

```

66         fail_total_count++;
67     end
68
69     // Check WREADY
70     if (tr.WREADY === 1'b0) begin
71         'uvm_info(get_type_name(), $sformatf("RESET CHECK | WREADY Expected=1'b0 Actual
72             =%0b | PASS", tr.WREADY), UVM_LOW)
73         pass_total_count++;
74     end else begin
75         'uvm_error(get_type_name(), $sformatf("RESET CHECK | WREADY Expected=1'b0 Actual
76             =%0b", tr.WREADY))
77         fail_total_count++;
78     end
79
80     // Check BVALID
81     if (tr.BVALID === 1'b0) begin
82         'uvm_info(get_type_name(), $sformatf("RESET CHECK | BVALID Expected=1'b0 Actual
83             =%0b | PASS", tr.BVALID), UVM_LOW)
84         pass_total_count++;
85     end else begin
86         'uvm_error(get_type_name(), $sformatf("RESET CHECK | BVALID Expected=1'b0 Actual
87             =%0b", tr.BVALID))
88         fail_total_count++;
89     end
90
91     // Check BRESP
92     if (tr.BRESP === 2'b00) begin
93         'uvm_info(get_type_name(), $sformatf("RESET CHECK | BRESP Expected=2'b00 Actual
94             =2'b%02b | PASS", tr.BRESP), UVM_LOW)
95         pass_total_count++;
96     end else begin
97         'uvm_error(get_type_name(), $sformatf("RESET CHECK | BRESP Expected=2'b00 Actual
98             =2'b%02b", tr.BRESP))
99         fail_total_count++;
100     end
101
102     // Check ARREADY
103     if (tr.ARREADY === 1'b1) begin
104         'uvm_info(get_type_name(), $sformatf("RESET CHECK | ARREADY Expected=1'b1 Actual
105             =%0b | PASS", tr.ARREADY), UVM_LOW)
106         pass_total_count++;
107     end else begin
108         'uvm_error(get_type_name(), $sformatf("RESET CHECK | ARREADY Expected=1'b1 Actual
109             =%0b", tr.ARREADY))
110         fail_total_count++;
111     end
112
113     // Check RVALID
114     if (tr.RVALID === 1'b0) begin
115         'uvm_info(get_type_name(), $sformatf("RESET CHECK | RVALID Expected=1'b0 Actual
116             =%0b | PASS", tr.RVALID), UVM_LOW)
117         pass_total_count++;
118     end else begin
119         'uvm_error(get_type_name(), $sformatf("RESET CHECK | RVALID Expected=1'b0 Actual
120             =%0b", tr.RVALID))
121         fail_total_count++;
122     end
123
124     // Check RRESP
125     if (tr.RRESP === 2'b00) begin
126         'uvm_info(get_type_name(), $sformatf("RESET CHECK | RRESP Expected=2'b00 Actual
127             =2'b%02b | PASS", tr.RRESP), UVM_LOW)
128         pass_total_count++;
129     end else begin
130         'uvm_error(get_type_name(), $sformatf("RESET CHECK | RRESP Expected=2'b00 Actual
131             =2'b%02b", tr.RRESP))
132         fail_total_count++;
133     end
134
135     // Check RDATA
136     if (tr.RDATA[0] === {DATA_WIDTH{1'b0}}) begin
137         'uvm_info(get_type_name(), $sformatf("RESET CHECK | RDATA Expected=0 Actual=%0h |
138             PASS", tr.RDATA[0]), UVM_LOW)
139         pass_total_count++;
140     end else begin
141         'uvm_error(get_type_name(), $sformatf("RESET CHECK | RDATA Expected=0 Actual=%0h"
142             , tr.RDATA[0]))
143         fail_total_count++;
144     end
145
146     // Check RLAST
147     if (tr.RLAST === 1'b0) begin
148         'uvm_info(get_type_name(), $sformatf("RESET CHECK | RLAST Expected=1'b0 Actual=%0
149             b | PASS", tr.RLAST), UVM_LOW)
150         pass_total_count++;
151     end else begin
152         'uvm_error(get_type_name(), $sformatf("RESET CHECK | RLAST Expected=1'b0 Actual
153             =%0b", tr.RLAST))
154         fail_total_count++;
155     end

```

```

139         end
140     end
141     else if(tr.op == WRITE)
142     begin
143         if (safeW) //Safe
144         begin
145             if(tr.WVALID && tr.WREADY && !tr.BVALID)
146             begin
147                 gmemory[(tr.AWADDR >> 2)+i] = tr.WDATA[i];
148                 'uvm_info(get_type_name(),$sformatf("WRITE DATA CHECK | Beat[%0d] | Addr=0x%0
149                     h | WDATA=0x%0h | PASS",i,(tr.AWADDR + i * (1 << tr.AWSIZE)),tr.WDATA[i])
150                     ,UVM_LOW)
151                 i= i+1;
152                 if(i == tr.AWLEN+1)
153                 begin
154                     'uvm_info(get_type_name(),$sformatf("WLAST CHECK | Expected=1 |
155                         Actual=%0b | Beat[%0d] | PASS",tr.WLAST,i),UVM_LOW)
156                     i = 0;
157                 end
158                 end
159                 else
160                 begin
161                     'uvm_error(get_type_name(),$sformatf("WLAST CHECK | Expected=1 |
162                         Actual=%0b | Beat[%0d] | FAILED",tr.WLAST,i))
163                     fail_total_count ++;
164                 end
165             end
166             if(tr.BVALID)
167             begin
168                 if(tr.BRESP == 0)
169                 begin
170                     'uvm_info(get_type_name(), $sformatf("BRESP CHECK | BRESP=2'b%02b (OKAY)
171                         | PASS", tr.BRESP), UVM_LOW)
172                     pass_total_count ++;
173                 end
174                 end
175                 else
176                 begin
177                     'uvm_error(get_type_name(), $sformatf("BRESP CHECK | BRESP=2'b%02b (OKAY)
178                         | ERROR", tr.BRESP))
179                     fail_total_count++;
180                 end
181             end
182         end
183     end
184     end
185     else
186     begin
187         if(tr.BVALID)
188         begin
189             if(tr.BRESP == 2'b10)
190             begin
191                 'uvm_info(get_type_name(),"BRESP ERROR IS RIGHT",UVM_LOW)
192                 pass_total_count ++;
193             end
194             end
195             else
196             begin
197                 'uvm_info(get_type_name(),"BRESP ERROR IS FAILED",UVM_LOW)
198                 fail_total_count++;
199             end
200         end
201     end
202     end
203     else if(tr.op == READ)
204     begin
205         if(safeR)
206         begin
207             if(tr.RVALID && tr.RREADY)
208             begin
209                 'uvm_info(get_type_name(),$sformatf("READ DATA CHECK | Beat[%0d] | Addr=0
210                     x%0h | RDATA=0x%0h | ",j,(tr.ARADDR + j * (1 << tr.ARSIZE)),tr.RDATA[
211                     j]),UVM_LOW)
212                 if(gmemory[(tr.ARADDR >> 2)+j] == tr.RDATA[j])
213                 'uvm_info(get_type_name(),$sformatf("READ DATA CHECK | Expected=%0h |
214                     Actual=%0h | Beat[%0d] | PASS",gmemory[(tr.ARADDR >> 2)+j],tr.RDATA[
215                     j],j),UVM_LOW)
216                 else
217                 begin
218                     'uvm_error(get_type_name(),$sformatf("READ DATA CHECK | Expected=%0h |
219                         Actual=%0h | Beat[%0d] | FAILED",gmemory[(tr.ARADDR >> 2)+j],tr.RDATA
220                         [j],j))
221                     fail_total_count ++;
222                 end
223                 end
224                 if(j == tr.ARLen+1)
225                 begin
226                     'uvm_info(get_type_name(),$sformatf("RLAST CHECK | Expected=1 |
227                         Actual=%0b | Beat[%0d] | PASS",tr.RLAST,j),UVM_LOW)
228                     j = 0;
229                 end
230                 end
231                 else
232                 begin
233                     'uvm_error(get_type_name(),$sformatf("RLAST CHECK | Expected=1 |

```

```

215         Actual=%0b | Beat[%0d] | FAILED",tr.RLAST,j))
216         fail_total_count ++;
217     end
218     if(tr.RVALID)
219     begin
220         if(tr.RRESP == 0)
221         begin
222             'uvm_info(get_type_name(), $sformatf("RRESP CHECK | RRESP=2'b%02b (
                OKAY) | PASS", tr.RRESP), UVM_LOW)
223             pass_total_count ++;
224         end
225         else
226         begin
227             'uvm_error(get_type_name(), $sformatf("RRESP CHECK | RRESP=2'b%02b (
                OKAY) | ERROR", tr.RRESP))
228             fail_total_count++;
229         end
230     end
231 end
232 else
233 begin
234     if(tr.RVALID)
235     begin
236         if(tr.RRESP == 2'b10)
237         begin
238             'uvm_info(get_type_name(),"RRESP ERROR IS RIGHT",UVM_LOW)
239             if(tr.RDATA[j] == 0)
240             begin
241                 'uvm_info(get_type_name(),$sformatf("READ DATA CHECK | Expected=%0h |
                Actual=%0h | Beat[%0d] | PASS",'h0,tr.RDATA[j],j),UVM_LOW)
242                 pass_total_count ++;
243             end
244             else
245             begin
246                 'uvm_error(get_type_name(),$sformatf("READ DATA CHECK | Expected=%0h
                | Actual=%0h | Beat[%0d] | FAILED",'h0,tr.RDATA[j],j))
247                 fail_total_count++;
248             end
249         end
250     end
251     else
252     begin
253         'uvm_info(get_type_name(),"RRESP ERROR IS FAILED",UVM_LOW)
254         fail_total_count++;
255     end
256 end
257 end
258 else if(tr.op == IDLE)
259 begin
260     // Check AWVALID remains deasserted
261     if (tr.AWVALID === 1'b0)
262     begin
263         'uvm_info(get_type_name(), "IDLE CHECK | AWVALID Expected=0 Actual=0 | PASS",
                UVM_LOW)
264         pass_total_count++;
265     end
266     else
267     begin
268         'uvm_error(get_type_name(), $sformatf("IDLE CHECK | AWVALID Expected=0 Actual=%0b
                ", tr.AWVALID))
269         fail_total_count++;
270     end
271 end
272 // Check ARVALID remains deasserted
273 if (tr.ARVALID === 1'b0)
274 begin
275     'uvm_info(get_type_name(), "IDLE CHECK | ARVALID Expected=0 Actual=0 | PASS",
                UVM_LOW)
276     pass_total_count++;
277 end
278 else
279 begin
280     'uvm_error(get_type_name(), $sformatf("IDLE CHECK | ARVALID Expected=0 Actual=%0b
                ", tr.ARVALID))
281     fail_total_count++;
282 end
283 // Check WVALID remains deasserted
284 if (tr.WVALID === 1'b0)
285 begin
286     'uvm_info(get_type_name(), "IDLE CHECK | WVALID Expected=0 Actual=0 | PASS",
                UVM_LOW)
287     pass_total_count++;
288 end
289 else
290 begin
291     'uvm_error(get_type_name(), $sformatf("IDLE CHECK | WVALID Expected=0 Actual=%0b

```

```

294         , tr.WVALID))
295         fail_total_count++;
296     end
297     // Check BVALID remains deasserted (no unexpected write response)
298     if (tr.BVALID === 1'b0)
299     begin
300         'uvm_info(get_type_name(), "IDLE CHECK | BVALID Expected=0 Actual=0 | PASS",
301             UVM_LOW)
302         pass_total_count++;
303     end
304     else
305     begin
306         'uvm_error(get_type_name(), $sformatf("IDLE CHECK | BVALID Expected=0 Actual=%0b"
307             , tr.BVALID))
308         fail_total_count++;
309     end
310     // Check RVALID remains deasserted (no unexpected read response)
311     if (tr.RVALID === 1'b0)
312     begin
313         'uvm_info(get_type_name(), "IDLE CHECK | RVALID Expected=0 Actual=0 | PASS",
314             UVM_LOW)
315         pass_total_count++;
316     end
317     else
318     begin
319         'uvm_error(get_type_name(), $sformatf("IDLE CHECK | RVALID Expected=0 Actual=%0b"
320             , tr.RVALID))
321         fail_total_count++;
322     end
323     end
324 endtask
325
326 task print_pass_summary();
327     'uvm_info(get_type_name(), "===== ", UVM_LOW)
328     'uvm_info(get_type_name(), "          SCOREBOARD PASS SUMMARY          ", UVM_LOW)
329     'uvm_info(get_type_name(), "===== ", UVM_LOW)
330     'uvm_info(get_type_name(), $sformatf("Total Checks Passed : %0d", pass_total_count), UVM_LOW)
331     'uvm_info(get_type_name(), $sformatf("Total Checks Failed : %0d", fail_total_count), UVM_LOW)
332     'uvm_info(get_type_name(), "===== ", UVM_LOW)
333 endtask
334
335 function void report_phase(uvm_phase phase);
336     super.report_phase(phase);
337     print_pass_summary();
338 endfunction
339
340 endclass
341 endpackage

```

Listing 5.1: AXI_SCB — AXI golden model checker.

```

1  package mem_scoreboard_pkg;
2      import uvm_pkg::*;
3      import parameters_pkg::*;
4      import mem_transaction_pkg::*;
5      'include "uvm_macros.svh"
6
7      class mem_scoreboard extends uvm_scoreboard;
8
9
10         logic [DATA_WIDTH-1:0] memory_golden [0:DEPTH-1];
11         logic [DATA_WIDTH-1:0] golden_rdata; // latch rdata form function
12         int pass_count = 0;
13         int fail_count = 0;
14
15         uvm_analysis_export #(mem_transaction) mem_analysis_export;
16         uvm_tlm_analysis_fifo #(mem_transaction) mem_fifo;
17         'uvm_component_utils(mem_scoreboard)
18
19         function new (string name = "mem_scoreboard" , uvm_component parent);
20             super.new(name, parent);
21             mem_analysis_export = new("mem_analysis_export",this);
22             mem_fifo = new("mem_fifo",this);
23         endfunction
24
25         function void build_phase (uvm_phase phase);
26             super.build_phase(phase);
27             'uvm_info(get_type_name(), "MEM SCOREBOARD BUILD PHASE", UVM_MEDIUM)
28             for (int i = 0; i < DEPTH; i++)
29             begin
30                 memory_golden[i] = '0;
31             end
32             golden_rdata = '0;
33         endfunction
34

```

```

35 function void connect_phase(uvm_phase phase);
36     mem_analysis_export.connect(mem_fifo.analysis_export);
37 endfunction
38
39 function void golden_mem(
40     input logic                rst_n,
41     input logic                mem_en,
42     input logic                mem_we,
43     input logic [ADDR_WIDTH-1:0] mem_addr,
44     input logic [DATA_WIDTH-1:0] mem_wdata,
45     output logic [DATA_WIDTH-1:0] mem_rdata
46 );
47     if (!rst_n)
48         begin
49             golden_rdata = '0;
50         end else if (mem_en)
51         begin
52             if (mem_we)
53                 memory_golden[mem_addr] = mem_wdata;
54             else
55                 golden_rdata = memory_golden[mem_addr];
56         end
57     mem_rdata = golden_rdata;
58 endfunction
59
60 task run_phase(uvm_phase phase);
61     mem_transaction mem_tr;
62     logic [DATA_WIDTH-1:0] exp_mem_rdata;
63     bit out_ok;
64
65     forever
66     begin
67         mem_fifo.get(mem_tr);
68
69         golden_mem(mem_tr.rst_n, mem_tr.mem_en, mem_tr.mem_we, mem_tr.mem_addr, mem_tr.mem_wdata,
70             exp_mem_rdata);
71
72         // 1. Reset Check
73         if (!mem_tr.rst_n)
74         begin
75             out_ok = (mem_tr.mem_rdata === exp_mem_rdata);
76             if (out_ok) begin
77                 'uvm_info(get_type_name(), $sformatf("PASS | OP=RESET ADDR=0x%03h | DUT: 0x%08h |
78                     EXP: 0x%08h",
79                         mem_tr.mem_addr, mem_tr.mem_rdata, exp_mem_rdata), UVM_LOW)
78                 pass_count++;
79             end else begin
80                 'uvm_error(get_type_name(), $sformatf("FAIL | OP=RESET ADDR=0x%03h | DUT: 0x%08h
81                     | EXP: 0x%08h",
82                         mem_tr.mem_addr, mem_tr.mem_rdata, exp_mem_rdata))
82                 fail_count++;
83             end
84         end
85         // 2. Memory Enabled Operations
86         else if (mem_tr.mem_en)
87         begin
88             // Write Transaction
89             if (mem_tr.mem_we)
90             begin
91                 if (memory_golden[mem_tr.mem_addr] === mem_tr.mem_wdata)
92                 begin
93                     'uvm_info(get_type_name(), $sformatf("PASS | OP=WRITE ADDR=0x%03h | DATA=0x
94                         %08h",
95                             mem_tr.mem_addr, mem_tr.mem_wdata), UVM_LOW)
95                     pass_count++;
96                 end
97                 else
98                 begin
99                     'uvm_error(get_type_name(), $sformatf("Failed | OP=Write ADDR=0x%03h | DUT: 0
100                         x%08h | EXP: 0x%08h",
101                             mem_tr.mem_addr, mem_tr.mem_wdata, memory_golden[mem_tr.mem_addr]))
101                     fail_count++;
102                 end
103             end
104             // Read Transaction Check
105             else
106             begin
107                 out_ok = (mem_tr.mem_rdata === exp_mem_rdata);
108                 if (out_ok)
109                 begin
110                     'uvm_info(get_type_name(), $sformatf("PASS | OP=READ ADDR=0x%03h | DUT: 0x%08
111                         h | EXP: 0x%08h",
112                             mem_tr.mem_addr, mem_tr.mem_rdata, exp_mem_rdata), UVM_LOW)
112                     pass_count++;
113                 end
114                 else
115                 begin
116                     'uvm_error(get_type_name(), $sformatf("FAIL | OP=READ ADDR=0x%03h | DUT: 0x
117                         %08h | EXP: 0x%08h",

```

```

117         mem_tr.mem_addr, mem_tr.mem_rdata, exp_mem_rdata))
118         fail_count++;
119     end
120 end
121 end
122 // 3. Memory Disabled / Idle Transaction
123 else
124 begin
125     'uvm_info(get_type_name(), "PASS | OP=IDLE (mem_en=0)", UVM_LOW)
126     pass_count++;
127 end
128 end
129 endtask
130
131 task print_pass_summary();
132     'uvm_info(get_type_name(), "===== ", UVM_LOW)
133     'uvm_info(get_type_name(), "          MEM SCOREBOARD PASS SUMMARY          ", UVM_LOW)
134     'uvm_info(get_type_name(), "===== ", UVM_LOW)
135     'uvm_info(get_type_name(), $sformatf("Total Checks Passed : %0d", pass_count), UVM_LOW)
136     'uvm_info(get_type_name(), $sformatf("Total Checks Failed : %0d", fail_count), UVM_LOW)
137     'uvm_info(get_type_name(), "===== ", UVM_LOW)
138 endtask
139
140 function void report_phase(uvm_phase phase);
141     super.report_phase(phase);
142     print_pass_summary();
143 endfunction
144
145 endclass
146
147 endpackage

```

Listing 5.2: mem_scb.sv — Raw memory golden model checker.

5.2 Functional Coverage Models

Two coverage subscribers (`axi_coverage` and `mem_coverage`) sample transaction attributes using TLM analysis FIFOs:

- **AXI Bus Covergroup (cg):** Analyzes address segments (`low_addr`, `mid_addr`, `high_addr`), burst lengths (`single`, `short`, `medium`, `long`), and cross coverage between transfer lengths, operations, and address modes.
- **Memory Covergroup (mem_cg):** Covers memory address ranges (`[0]`, `[1:255]`, `[256:511]`, `[512:767]`, `[768:1023]`) and operation direction (`read`, `write`).

```

1 package axi_coverage_pkg;
2
3     import axi_transaction_pkg::*;
4     import uvm_pkg::*;
5
6     'include "uvm_macros.svh"
7
8
9     class axi_coverage extends uvm_component;
10
11         uvm_analysis_export #(axi_transaction) analysis_export;
12         uvm_tlm_analysis_fifo #(axi_transaction) fifo;
13         'uvm_component_utils(axi_coverage)
14
15         axi_transaction tr;
16         covergroup cg ;
17         //option.per_instance = 1;
18         cp_awaddr: coverpoint tr.AWADDR {
19             bins low_addr = {[0 : 1023]};
20             bins mid_addr = {[1024 : 3071]};
21             bins high_addr = {[3072 : 4095]};
22         }
23
24         cp_awlen: coverpoint tr.AWLEN {
25             bins len_single = {0}; // Single beat
26             bins len_short = {[1 : 15]}; // Short burst
27             bins len_med = {[16 : 127]}; // Medium burst
28             bins len_max = {[128 : 255]}; // Long burst
29         }
30
31         cp_araddr: coverpoint tr.ARADDR {
32             bins low_addr = {[0 : 1023]};
33             bins mid_addr = {[1024 : 3071]};
34             bins high_addr = {[3072 : 4095]};
35         }
36
37         cp_arlen: coverpoint tr.ARLLEN {
38             bins len_single = {0};

```

```

39     bins len_short = {[1 : 15]};
40     bins len_med   = {[16 : 127]};
41     bins len_max   = {[128 : 255]};
42 }
43
44 cp_awvalid: coverpoint tr.AWVALID;
45 cp_arvalid: coverpoint tr.ARVALID;
46 cp_op: coverpoint tr.op;
47 cp_address_mode: coverpoint tr.addr_mode {
48     bins mode_0 = {0};
49     bins mode_1 = {1};
50     bins mode_2 = {2};
51 }
52 cp_op_mode: cross cp_op, cp_address_mode;
53 cp_wadd_len: cross cp_awaddr, cp_awlen {
54     bins low_singlelen = binsof(cp_awaddr.low_addr) && binsof(cp_awlen.len_single);
55 }
56 cp_radd_len: cross cp_araddr, cp_arlen {
57     bins low_singlelen = binsof(cp_araddr.low_addr) && binsof(cp_arlen.len_single);
58 }
59
60 endgroup
61
62 function new (string name = "axi_coverage" , uvm_component parent);
63     super.new(name, parent);
64     cg = new();
65     analysis_export = new("analysis_export",this);
66     fifo = new("fifo",this);
67 endfunction
68
69 function void build_phase (uvm_phase phase);
70     super.build_phase(phase);
71     'uvm_info(get_type_name(), "AXI COVERAGE BUILD PHASE", UVM_MEDIUM)
72 endfunction
73
74 function void connect_phase(uvm_phase phase);
75     analysis_export.connect(fifo.analysis_export);
76 endfunction
77
78 task run_phase(uvm_phase phase);
79
80     forever begin
81         fifo.get(tr);
82         'uvm_info(get_type_name(), "Coverage sampled transaction!", UVM_LOW)
83         cg.sample();
84     end
85 endtask
86
87 endclass
88 endpackage

```

Listing 5.3: AXI Coverage.

```

1  package mem_coverage_pkg;
2
3  import mem_transaction_pkg::*;
4  import uvm_pkg::*;
5
6  'include "uvm_macros.svh"
7
8
9  class mem_coverage extends uvm_component;
10
11     uvm_analysis_export #(mem_transaction) mem_analysis_export;
12     uvm_tlm_analysis_fifo #(mem_transaction) mem_fifo;
13     'uvm_component_utils(mem_coverage)
14
15     mem_transaction mem_tr;
16     covergroup mem_cg ;
17         option.per_instance = 1;
18         addr_cp: coverpoint mem_tr.mem_addr {
19             bins first = {0};
20             bins low = {[1:255]};
21             bins mid_low = {[256:511]};
22             bins mid_high = {[512:767]};
23             bins high = {[768:1023]};
24         }
25         we_cp: coverpoint mem_tr.mem_we {
26             bins write = {1};
27             bins read = {0};
28         }
29     endgroup
30
31     function new (string name = "mem_coverage" , uvm_component parent);
32         super.new(name, parent);
33         mem_cg = new();
34         mem_analysis_export = new("mem_analysis_export",this);
35         mem_fifo = new("mem_fifo",this);

```

```

36     endfunction
37
38     function void build_phase (uvm_phase phase);
39         super.build_phase(phase);
40         'uvm_info(get_type_name(), "MEM COVERAGE BUILD PHASE", UVM_MEDIUM)
41     endfunction
42
43     function void connect_phase(uvm_phase phase);
44         mem_analysis_export.connect(mem_fifo.analysis_export);
45     endfunction
46
47     task run_phase(uvm_phase phase);
48
49         forever begin
50             mem_fifo.get(mem_tr);
51             'uvm_info(get_type_name(), "MEM Coverage sampled transaction!", UVM_LOW)
52             mem_cg.sample();
53         end
54     endtask
55
56 endclass
57 endpackage

```

Listing 5.4: Memory Coverage.

5.3 Env, Test and Top

```

1  package axi_env_pkg;
2
3
4      import uvm_pkg::*;
5      import axi_agent_pkg::*;
6      import mem_agent_pkg::*;
7      import axi_scoreboard_pkg::*;
8      import mem_scoreboard_pkg::*;
9      import axi_coverage_pkg::*;
10     import common_cfg_pkg::*;
11     import mem_coverage_pkg::*;
12     'include "uvm_macros.svh"
13
14     class axi_env extends uvm_env;
15         common_cfg m_cfg;
16         axi_agent agt;
17         mem_agent mem_agt;
18         axi_scoreboard scb;
19         mem_scoreboard mem_scb;
20         axi_coverage cov;
21         mem_coverage mem_cov;
22         uvm_active_passive_enum is_active = UVM_ACTIVE;
23
24         'uvm_component_utils(axi_env)
25
26     function new (string name = "axi_env" , uvm_component parent);
27         super.new(name, parent);
28     endfunction
29
30     function void build_phase (uvm_phase phase);
31         super.build_phase(phase);
32         agt = axi_agent::type_id::create("agt", this);
33         scb = axi_scoreboard::type_id::create("scb", this);
34         cov = axi_coverage::type_id::create("cov", this);
35         mem_cov = mem_coverage::type_id::create("mem_cov", this);
36         mem_agt = mem_agent::type_id::create("mem_agt", this);
37         mem_scb = mem_scoreboard::type_id::create("mem_scb", this);
38
39         if(!(uvm_config_db #(uvm_active_passive_enum)::get(this, "", "is_active", is_active)))
40             'uvm_fatal(get_type_name(), "FAILED TO GET UVM ACTIVE PASSIVE AGENT")
41
42         'uvm_info(get_type_name(), "AXI ENV BUILD PHASE", UVM_MEDIUM)
43         uvm_config_db #(common_cfg)::get(this, "", "m_cfg", m_cfg);
44     endfunction
45
46     function void connect_phase(uvm_phase phase);
47         super.connect_phase(phase);
48         if(is_active == UVM_ACTIVE)
49             begin
50                 agt.drv.m_cfg=m_cfg;
51             end
52             agt.mon.m_cfg=m_cfg;
53             agt.mon.ap.connect(scb.analysis_export);
54             agt.mon.ap.connect(cov.analysis_export);
55             mem_agt.mem_mon.mem_ap.connect(mem_scb.mem_analysis_export);
56             mem_agt.mem_mon.mem_ap.connect(mem_cov.mem_analysis_export);
57             'uvm_info(get_type_name(), "AXI ENV CONNECT PHASE", UVM_MEDIUM)
58     endfunction
59

```

```

60     endclass
61 endpackage

```

Listing 5.5: Axi Env.

```

1  package axi_test_pkg;
2
3      import uvm_pkg::*;
4      import axi_env_pkg::*;
5      import axi_sequence_pkg::*;
6      import axi_driver_pkg::*;
7      import axi_driver_delay_pkg::*;
8      import common_cfg_pkg::*;
9      'include "uvm_macros.svh"
10
11     class axi_test extends uvm_test;
12
13         axi_env env;
14         axi_sequence seq;
15         common_cfg m_cfg;
16         'uvm_component_utils(axi_test)
17
18         function new (string name = "axi_test" , uvm_component parent);
19             super.new(name, parent);
20         endfunction
21
22         function void build_phase (uvm_phase phase);
23
24             //uvm_factory::get().set_type_override_by_type(axi_driver::get_type(), axi_driver_delay::
25                 get_type());
26             super.build_phase(phase);
27             env = axi_env::type_id::create("env", this);
28             seq = axi_sequence::type_id::create("seq");
29             m_cfg = common_cfg::type_id::create("m_cfg");
30             'uvm_info(get_type_name(), "AXI TEST BUILD PHASE", UVM_LOW)
31
32             uvm_config_db #(common_cfg)::set(this, "*", "m_cfg", m_cfg);
33         endfunction
34
35         task run_phase(uvm_phase phase);
36             phase.raise_objection(this);
37             fork
38                 begin
39                     seq.start(env.agt.sqr);
40                     #1000ns;
41                 end
42             join
43             begin
44                 'uvm_info(get_type_name(), "Starting AXI SEQUENCE", UVM_LOW)
45             end
46             phase.drop_objection(this);
47         endtask
48
49         function void end_of_elaboration_phase(uvm_phase phase);
50             super.end_of_elaboration_phase(phase);
51             uvm_top.print_topology();
52         endfunction
53     endclass
54 endpackage

```

Listing 5.6: Axi Test.

```

1  'include "uvm_macros.svh"
2  import uvm_pkg::*;
3  import axi_test_pkg::*;
4
5  bind axi4_memory mem_if #(
6      .ADDR_WIDTH(10),
7      .DATA_WIDTH(32)
8  ) mem_bd_if (
9      .clk      (clk),
10     .rst_n    (rst_n),
11     .mem_en   (mem_en),
12     .mem_we   (mem_we),
13     .mem_addr (mem_addr),
14     .mem_wdata(mem_wdata),
15     .mem_rdata(mem_rdata)
16 );
17
18 module top;
19
20     AXI_if axi_vif();
21     axi4 dut (
22         .ACLK      (axi_vif.ACLK),
23         .ARESETn   (axi_vif.ARESETn),
24

```

```

25      // Write address channel
26      .AWADDR  (axi_vif.AWADDR),
27      .AWLEN   (axi_vif.AWLEN),
28      .AWSIZE  (axi_vif.AWSIZE),
29      .AWVALID (axi_vif.AWVALID),
30      .AWREADY (axi_vif.AWREADY),
31
32      // Write data channel
33      .WDATA   (axi_vif.WDATA),
34      .WVALID  (axi_vif.WVALID),
35      .WLAST   (axi_vif.WLAST),
36      .WREADY  (axi_vif.WREADY),
37
38      // Write response channel
39      .BRESP   (axi_vif.BRESP),
40      .BVALID  (axi_vif.BVALID),
41      .BREADY  (axi_vif.BREADY),
42
43      // Read address channel
44      .ARADDR  (axi_vif.ARADDR),
45      .ARLEN   (axi_vif.ARLEN),
46      .ARSIZE  (axi_vif.ARSIZE),
47      .ARVALID (axi_vif.ARVALID),
48      .ARREADY (axi_vif.ARREADY),
49
50      // Read data channel
51      .RDATA   (axi_vif.RDATA),
52      .RRESP   (axi_vif.RRESP),
53      .RVALID  (axi_vif.RVALID),
54      .RLAST   (axi_vif.RLAST),
55      .RREADY  (axi_vif.RREADY)
56  );
57
58  initial begin
59      axi_vif.ACLK = 0;
60      forever
61      begin
62          #2ns axi_vif.ACLK = ~axi_vif.ACLK;
63      end
64  end
65
66  initial begin
67      uvm_config_db #(uvm_active_passive_enum)::set(null, "uvm_test_top.*", "is_active", UVM_ACTIVE);
68      uvm_config_db #(virtual AXI_if)::set(null, "uvm_test_top.env.agt.*", "vif", axi_vif);
69      uvm_config_db #(virtual mem_if)::set(null, "uvm_test_top.env.mem.agt.*", "mem_vif", top.dut.mem_inst
        .mem_bd_if);
70      uvm_top.set_report_verbosity_level(UVM_LOW);
71      run_test ("axi_test");
72  end
73
74  endmodule

```

Listing 5.7: Axi Top.

CHAPTER 6

SystemVerilog Assertions (SVA)

6.1 Bus Invariants and Check Properties

Concurrent SVA properties running inside AXI_if enforce protocol rules during simulation:

Table 6.1: SystemVerilog Assertions Suite implemented in AXI_properties_pkg.

Property Identifier	Formal Behavioral Invariant	Scope
p_bvalid_after_wlast	\$rose(BVALID) -> \$past(WVALID && WREADY && WLAST)	Write Handshake
RREADY_after_wlast_is_zero	(RLAST && RREADY && RVALID) -> ##1 (!RREADY)	Read Handshake
awready_after_awvalid	AWVALID -> ##1 (!AWREADY)	Address Timing
WLAST_AND_RLAST	not (WLAST && RLAST)	Mutual Exclusion
WLAST_AND_AWREADY	not (WLAST && AWREADY)	Mutual Exclusion
RLAST_AND_ARREADY	not (RLAST && ARREADY)	Mutual Exclusion
WLAST_AND_AWVALID	not (WLAST && AWVALID)	Mutual Exclusion
RLAST_AND_ARVALID	not (RLAST && ARVALID)	Mutual Exclusion

```
1 package AXI_properties_pkg;
2   property p_bvalid_after_wlast(ACLK, ARESETn, BVALID, WVALID, WREADY, WLAST);
3     @(posedge ACLK) disable iff (!ARESETn)
4       $rose(BVALID) |-> $past(WVALID && WREADY && WLAST);
5   endproperty
6
7   property RREADY_after_wlast_is_zero(ACLK, ARESETn, RLAST, RREADY, RVALID);
8     @(posedge ACLK) disable iff (!ARESETn)
9       (RLAST && RREADY && RVALID) |-> ##1 (!RREADY);
10  endproperty
11
12  property awready_after_awvalid(ACLK, ARESETn, AWVALID, AWREADY);
13    @(posedge ACLK) disable iff (!ARESETn)
14      AWVALID |-> ##1 (!AWREADY);
15  endproperty
16
17  property WLAST_AND_RLAST(ACLK, ARESETn, WLAST, RLAST);
18    @(posedge ACLK) disable iff (!ARESETn)
19      not (WLAST && RLAST);
20  endproperty
21 endpackage
```

Listing 6.1: AXI_properties_pkg.sv — SVA formal properties.

CHAPTER 7

Factory Type Override Demonstration

7.1 Factory Polymorphism Configuration

The testbench uses factory type override to swap `axi_driver` with `axi_driver_delay` without modifying the underlying sequence code:

```
# ** Report counts by severity
# UVM_INFO :1301419
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [MEM_MON] 361980
# [MON] 173830
# [Questa UVM] 2
# [RNTST] 1
# [TEST_DONE] 1
# [UVMTOP] 1
# [axi_coverage] 87016
# [axi_driver_delay] 101
# [axi_monitor] 2002
# [axi_scoreboard] 314495
# [axi_test] 2
# [mem_coverage] 180990
# [mem_monitor] 2
# [mem_scoreboard] 180996
# ** Note: $finish : C:/questasim64_2021.1/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 723960 ns Iteration: 54 Instance: /top
```

Figure 7.1: Axi driver delay.

```
1 function void axi_test::build_phase(uvm_phase phase);
2     // Factory Type Override applied before creating hierarchy:
3     uvm_factory::get().set_type_override_by_type(
4         axi_driver::get_type(),
5         axi_driver_delay::get_type()
6     );
7     super.build_phase(phase);
8     env = axi_env::type_id::create("env", this);
9 endfunction
```

Listing 7.1: Factory override invocation in test build phase.

7.2 Topology Verification Output

Calling `uvm_top.print_topology()` inside `end_of_elaboration_phase` confirms the driver instantiation changes dynamically:

Before Override (Default)		After Type Override Applied	
uvm_test_top	axi_test	uvm_test_top	axi_test
env	axi_env	env	axi_env
agt	axi_agent	agt	axi_agent
sqr	axi_sequencer	sqr	axi_sequencer
drv	axi_driver	drv	axi_driver_delay
mon	axi_monitor	mon	axi_monitor
cov	axi_coverage	cov	axi_coverage
scb	axi_scoreboard	scb	axi_scoreboard
mem_agt	mem_agent	mem_agt	mem_agent
mem_mon	mem_monitor	mem_mon	mem_monitor
mem_cov	mem_coverage	mem_cov	mem_coverage
mem_scb	mem_scoreboard	mem_scb	mem_scoreboard

Figure 7.2: Simulator topology printout before vs. after factory type override.

CHAPTER 8

Testing Results & Waveform Verification

Functional verification of the AMBA AXI4-compliant memory-mapped slave controller and synchronous single-port SRAM was conducted across targeted operations, bus quiescent cycles, 4KB boundary crossings, and constrained-random regressions. All verification checks were monitored concurrently by the active AXI bus agent and the passive memory agent.

8.1 System Reset Verification

During the initialization phase, **ARESETn** is asserted active-low. The cycle-by-cycle scoreboards verify interface quiescence, pipeline flushing, and output register clearing.

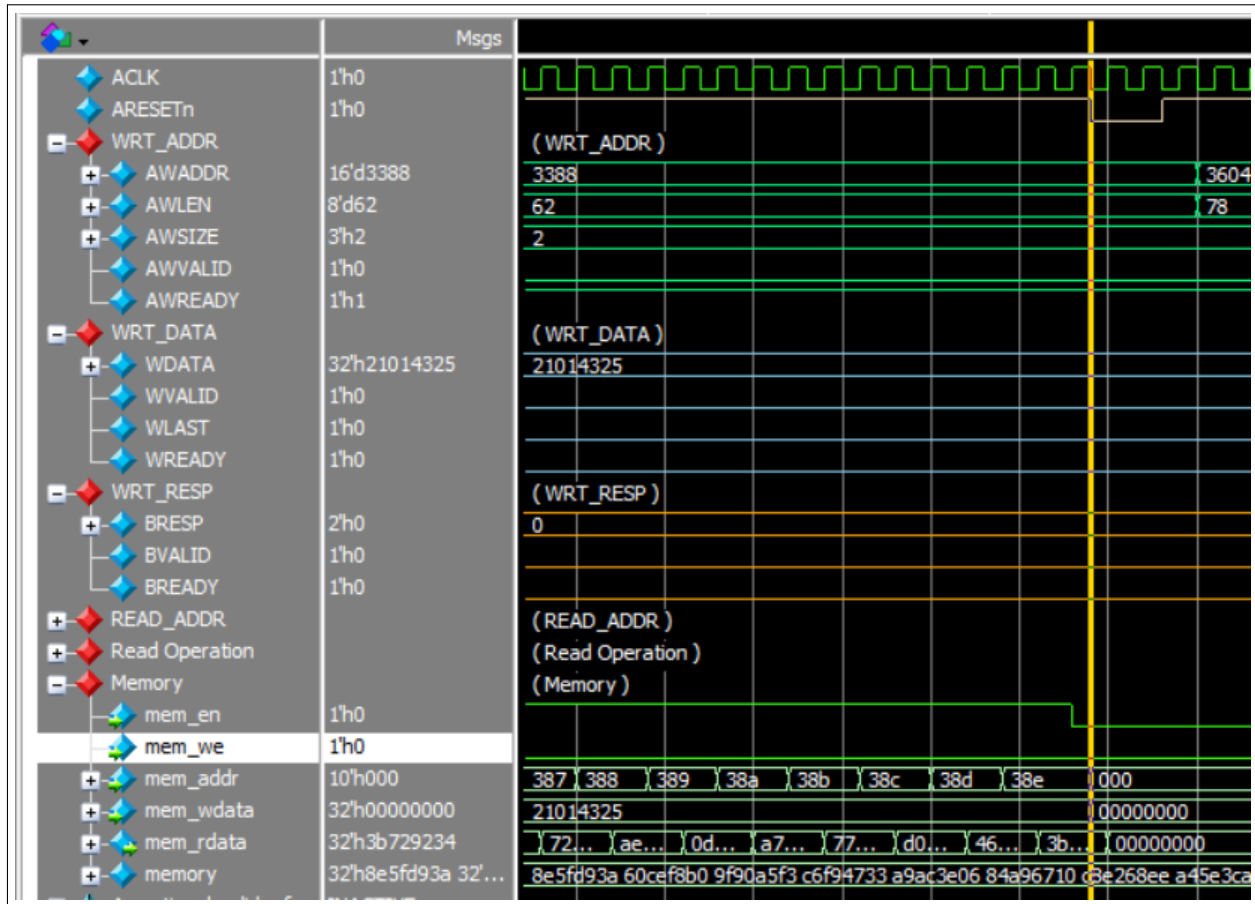


Figure 8.1: Waveform execution showing **ARESETn** assertion, bus quiescence, and clean initialization.

Behavioral Analysis:

- While **ARESETn** remains low, all channel handshakes and valid flags (**WVALID**, **BVALID**, **RVALID**, **RLAST**) are forced to 1'b0.
- Both **AWREADY** and **ARREADY** assert high, confirming readiness to accept address phase transfers upon reset deassertion.
- Internal memory control lines (**mem_en**, **mem_we**) remain low, and internal memory address and data registers clear to '0.

8.2 Burst Write Transaction Verification (Write operation)

Multi-beat write transactions verify channel synchronization across the write address, write data, and write response channels[cite: 1].

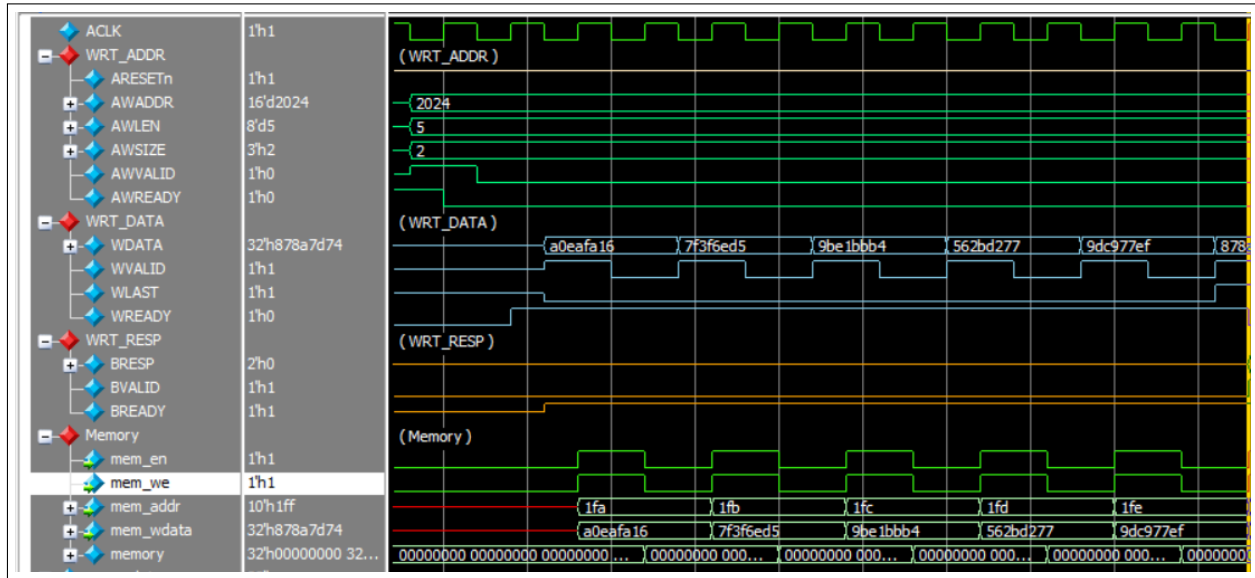


Figure 8.2: Multi-beat write burst to address 2024 (AWLEN = 5) with internal RAM writing (0x1fa to 0x1fe).

Behavioral Analysis:

- **Address Phase:** The write address phase samples $AWADDR = 2024$, $AWLEN = 5$ (6 transfer beats total), and $AWSIZE = 2$ (4 bytes per word beat)[cite: 1].
- **Data Handshake:** Data payloads (32'h878a7d74, 32'h878a7d74, 32'h878a7d74, 32'h878a7d74, 32'h878a7d74, and 32'h878a7d74) are accepted over $WDATA$ upon concurrent assertion of $WVALID$ and $WREADY$ [cite: 1].
- **Internal Memory Mapping:** The internal word address steps sequentially through locations 10'h1fa, 10'h1fb, 10'h1fc, 10'h1fd, 10'h1fe, and 10'h1ff ($mem_addr = AWADDR \gg 2$) with $mem_we = 1$ and $mem_en = 1$.
- **Response Phase:** The driver asserts $WLAST = 1$ on the final beat, after which the slave asserts $BVALID = 1$ returning $BRESP = 2'b00$ (OKAY)[cite: 1].

8.2.1 UVM Simulation and Scoreboard Execution Logs

Figures 8.3 and 8.4 capture the live execution transcript from both the active AXI bus scoreboard (`axi_scoreboard`) and the passive memory scoreboard (`mem_scoreboard`), tracking cycle-by-cycle beat validation through simulation completion:

```

# UVM_INFO a_scoreboard.sv(148) @ 18: uvm_test_top.env.scb [axi_scoreboard] WRITE DATA CHECK | Beat[0] | Addr=0x7e8 | WDATA=0xa0eaf16 | PASS
# UVM_INFO a_scoreboard.sv(49) @ 18: uvm_test_top.env.scb [axi_scoreboard] Waiting for transaction
# UVM_INFO mem_coverage.sv(51) @ 18: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(125) @ 18: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=IDLE (mem_en=0)
# UVM_INFO mem_monitor.sv(44) @ 22: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 22: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_coverage.sv(51) @ 22: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(125) @ 22: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=IDLE (mem_en=0)
# UVM_INFO a_monitor.sv(91) @ 26: uvm_test_top.env.agt.mon [MON] About to send transaction Write OP
# UVM_INFO a_monitor.sv(93) @ 26: uvm_test_top.env.agt.mon [MON] Transaction sent Write OP
# UVM_INFO mem_monitor.sv(44) @ 26: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 26: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO a_coverage.sv(82) @ 26: uvm_test_top.env.cov [axi_coverage] Coverage sampled transaction!
# UVM_INFO a_scoreboard.sv(53) @ 26: uvm_test_top.env.scb [axi_scoreboard] Transaction received
# -----
# UVM_INFO a_scoreboard.sv(148) @ 26: uvm_test_top.env.scb [axi_scoreboard] WRITE DATA CHECK | Beat[1] | Addr=0x7ec | WDATA=0x7f3f6ed5 | PASS
# UVM_INFO a_scoreboard.sv(49) @ 26: uvm_test_top.env.scb [axi_scoreboard] Waiting for transaction
# UVM_INFO mem_coverage.sv(51) @ 26: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(93) @ 26: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=WRITE ADDR=0x1fa | DATA=0xa0eaf16
# UVM_INFO mem_monitor.sv(44) @ 30: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 30: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_coverage.sv(51) @ 30: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(125) @ 30: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=IDLE (mem_en=0)
# UVM_INFO a_monitor.sv(91) @ 34: uvm_test_top.env.agt.mon [MON] About to send transaction Write OP
# UVM_INFO a_monitor.sv(93) @ 34: uvm_test_top.env.agt.mon [MON] Transaction sent Write OP
# UVM_INFO mem_monitor.sv(44) @ 34: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 34: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO a_coverage.sv(82) @ 34: uvm_test_top.env.cov [axi_coverage] Coverage sampled transaction!
# UVM_INFO a_scoreboard.sv(53) @ 34: uvm_test_top.env.scb [axi_scoreboard] Transaction received
# -----
# UVM_INFO a_scoreboard.sv(148) @ 34: uvm_test_top.env.scb [axi_scoreboard] WRITE DATA CHECK | Beat[2] | Addr=0x7f0 | WDATA=0x9be1bbb4 | PASS
# UVM_INFO a_scoreboard.sv(49) @ 34: uvm_test_top.env.scb [axi_scoreboard] Waiting for transaction
# UVM_INFO mem_coverage.sv(51) @ 34: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(93) @ 34: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=WRITE ADDR=0x1fb | DATA=0x7f3f6ed5
# UVM_INFO mem_monitor.sv(44) @ 38: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 38: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_coverage.sv(51) @ 38: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(125) @ 38: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=IDLE (mem_en=0)
# UVM_INFO a_monitor.sv(91) @ 42: uvm_test_top.env.agt.mon [MON] About to send transaction Write OP
# UVM_INFO a_monitor.sv(93) @ 42: uvm_test_top.env.agt.mon [MON] Transaction sent Write OP
# UVM_INFO mem_monitor.sv(44) @ 42: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 42: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO a_coverage.sv(82) @ 42: uvm_test_top.env.cov [axi_coverage] Coverage sampled transaction!
# UVM_INFO a_scoreboard.sv(53) @ 42: uvm_test_top.env.scb [axi_scoreboard] Transaction received
# -----
# UVM_INFO a_scoreboard.sv(148) @ 42: uvm_test_top.env.scb [axi_scoreboard] WRITE DATA CHECK | Beat[3] | Addr=0x7f4 | WDATA=0x562bd277 | PASS
# UVM_INFO a_scoreboard.sv(49) @ 42: uvm_test_top.env.scb [axi_scoreboard] Waiting for transaction
# UVM_INFO mem_coverage.sv(51) @ 42: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(93) @ 42: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=WRITE ADDR=0x1fc | DATA=0x9be1bbb4
# UVM_INFO mem_monitor.sv(44) @ 46: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 46: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_coverage.sv(51) @ 46: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(125) @ 46: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=IDLE (mem_en=0)
# UVM_INFO a_monitor.sv(91) @ 50: uvm_test_top.env.agt.mon [MON] About to send transaction Write OP
# UVM_INFO a_monitor.sv(93) @ 50: uvm_test_top.env.agt.mon [MON] Transaction sent Write OP
# UVM_INFO mem_monitor.sv(44) @ 50: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 50: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO a_coverage.sv(82) @ 50: uvm_test_top.env.cov [axi_coverage] Coverage sampled transaction!
# UVM_INFO a_scoreboard.sv(53) @ 50: uvm_test_top.env.scb [axi_scoreboard] Transaction received
# -----

```

Figure 8.3: UVM scoreboard execution transcript for write beats 0 through 3.

```

# -----
# UVM_INFO a_scoreboard.sv(148) @ 50: uvm_test_top.env.scb [axi_scoreboard] WRITE DATA CHECK | Beat[4] | Addr=0x7f8 | WDATA=0x9dc977ef | PASS
# UVM_INFO a_scoreboard.sv(49) @ 50: uvm_test_top.env.scb [axi_scoreboard] Waiting for transaction
# UVM_INFO mem_coverage.sv(51) @ 50: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(93) @ 50: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=WRITE ADDR=0x1fd | DATA=0x562bd277
# UVM_INFO mem_monitor.sv(44) @ 54: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 54: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_coverage.sv(51) @ 54: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(125) @ 54: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=IDLE (mem_en=0)
# UVM_INFO a_monitor.sv(91) @ 58: uvm_test_top.env.agt.mon [MON] About to send transaction Write OP
# UVM_INFO a_monitor.sv(93) @ 58: uvm_test_top.env.agt.mon [MON] Transaction sent Write OP
# UVM_INFO mem_monitor.sv(44) @ 58: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 58: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO a_coverage.sv(82) @ 58: uvm_test_top.env.cov [axi_coverage] Coverage sampled transaction!
# UVM_INFO a_scoreboard.sv(53) @ 58: uvm_test_top.env.scb [axi_scoreboard] Transaction received
# -----
# Name      Type      Size  Value
# -----
# -----
# UVM_INFO a_scoreboard.sv(148) @ 58: uvm_test_top.env.scb [axi_scoreboard] WRITE DATA CHECK | Beat[5] | Addr=0x7fc | WDATA=0x878a7d74 | PASS
# UVM_INFO a_scoreboard.sv(153) @ 58: uvm_test_top.env.scb [axi_scoreboard] WLAST CHECK | Expected=1 | Actual=1 | Beat[6] | PASS
# UVM_INFO a_scoreboard.sv(49) @ 58: uvm_test_top.env.scb [axi_scoreboard] Waiting for transaction
# UVM_INFO mem_coverage.sv(51) @ 58: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(93) @ 58: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=WRITE ADDR=0x1fe | DATA=0x9dc977ef
# UVM_INFO mem_monitor.sv(44) @ 62: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 62: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_coverage.sv(51) @ 62: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(125) @ 62: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=IDLE (mem_en=0)
# UVM_INFO a_monitor.sv(101) @ 66: uvm_test_top.env.agt.mon [MON] About to send transaction
# UVM_INFO a_monitor.sv(103) @ 66: uvm_test_top.env.agt.mon [MON] Transaction sent
# UVM_INFO mem_monitor.sv(44) @ 66: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO mem_monitor.sv(46) @ 66: uvm_test_top.env.mem_agt.mem_mon [MEM_MON] About to send MEM transaction
# UVM_INFO a_coverage.sv(82) @ 66: uvm_test_top.env.cov [axi_coverage] Coverage sampled transaction!
# UVM_INFO a_scoreboard.sv(53) @ 66: uvm_test_top.env.scb [axi_scoreboard] Transaction received
# -----
# -----
# UVM_INFO a_scoreboard.sv(166) @ 66: uvm_test_top.env.scb [axi_scoreboard] BRESP CHECK | BRESP=2'b00 (OKAY) | PASS
# UVM_INFO a_scoreboard.sv(49) @ 66: uvm_test_top.env.scb [axi_scoreboard] Waiting for transaction
# UVM_INFO mem_coverage.sv(51) @ 66: uvm_test_top.env.mem_cov [mem_coverage] MEM Coverage sampled transaction!
# UVM_INFO mem_scb.sv(93) @ 66: uvm_test_top.env.mem_scb [mem_scoreboard] PASS | OP=WRITE ADDR=0x1ff | DATA=0x878a7d74
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 1064: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
# UVM_INFO mem_scb.sv(132) @ 1064: uvm_test_top.env.mem_scb [mem_scoreboard] =====
# UVM_INFO mem_scb.sv(133) @ 1064: uvm_test_top.env.mem_scb [mem_scoreboard] MEM SCOREBOARD PASS SUMMARY
# UVM_INFO mem_scb.sv(134) @ 1064: uvm_test_top.env.mem_scb [mem_scoreboard] =====
# UVM_INFO mem_scb.sv(135) @ 1064: uvm_test_top.env.mem_scb [mem_scoreboard] Total Checks Passed : 266
# UVM_INFO mem_scb.sv(136) @ 1064: uvm_test_top.env.mem_scb [mem_scoreboard] Total Checks Failed : 0
# UVM_INFO mem_scb.sv(137) @ 1064: uvm_test_top.env.mem_scb [mem_scoreboard] =====
# UVM_INFO a_scoreboard.sv(325) @ 1064: uvm_test_top.env.scb [axi_scoreboard] =====
# UVM_INFO a_scoreboard.sv(326) @ 1064: uvm_test_top.env.scb [axi_scoreboard] SCOREBOARD PASS SUMMARY
# UVM_INFO a_scoreboard.sv(327) @ 1064: uvm_test_top.env.scb [axi_scoreboard] =====
# UVM_INFO a_scoreboard.sv(328) @ 1064: uvm_test_top.env.scb [axi_scoreboard] Total Checks Passed : 1
# UVM_INFO a_scoreboard.sv(329) @ 1064: uvm_test_top.env.scb [axi_scoreboard] Total Checks Failed : 0
# UVM_INFO a_scoreboard.sv(330) @ 1064: uvm_test_top.env.scb [axi_scoreboard] =====
#
# --- UVM Report Summary ---

```

Figure 8.4: Write beats 4–5, WLAST check, BRESP validation, and scoreboard pass summary.

Log Analysis:

- **Beat-by-Beat Check:** `axi_scoreboard` tracks every transfer on the bus from Beat[0] at address 0x7e8 through Beat[5] at address 0x7fc, logging sequential data matches against golden packets.
- **Internal Memory Verification:** In parallel, `mem_scoreboard` captures write access at internal word addresses (0x1fa through 0x1ff), verifying exact data alignment between the bus payload and SRAM storage.
- **WLAST and BRESP Validation:** At time 58, WLAST assertion evaluates with `Expected=1` and `Actual=1`. At time 66, the write response handshake finishes with `BRESP=2'b00 (OKAY)`.
- **Test Completion:** The end-of-test summary confirms 266 checks passed (0 failures) in `mem_scoreboard` and 1 targeted burst check passed (0 failures) in `axi_scoreboard`.

8.3 Write-to-Read Verification (Read Operation)

A direct read burst following the write burst verifies that written data is accurately retained and retrievable from the internal synchronous memory array[cite: 1].

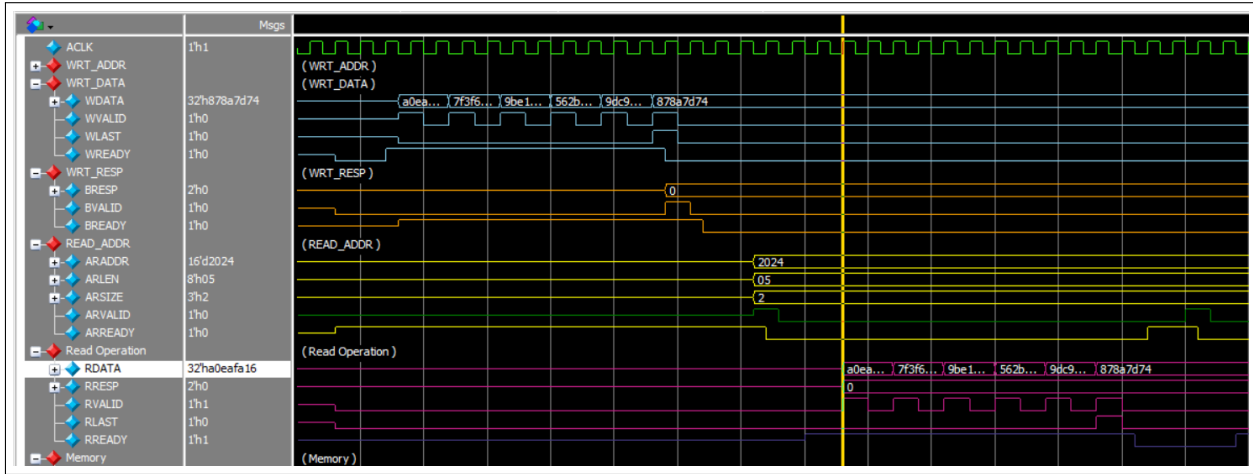


Figure 8.5: Write followed by read-back check targeting address 2024, confirming complete payload matching.

Behavioral Analysis:

- Following write completion, the master initiates a read burst with $ARADDR = 2024$, $ARLEN = 5$, and $ARSIZE = 2$.
- The slave reads memory locations $10'h1fa$ through $10'h1ff$, presenting data sequentially on $RDATA$ ($32'h878a7d74$, $32'h7f3f6ed5$, ..., $32'h878a7d74$) alongside $RVALID = 1$.
- On the terminal beat, $RLAST = 1$ asserts synchronously with $RRESP = 2'b00$ (OKAY), matching golden model scoreboard records with zero latency skews.

8.4 Quiescent Bus Verification ($AWVALID = 0$, $ARVALID = 0$)

Verifying idle bus states confirms that deasserted control lines prevent spurious memory access and unmapped slave responses.

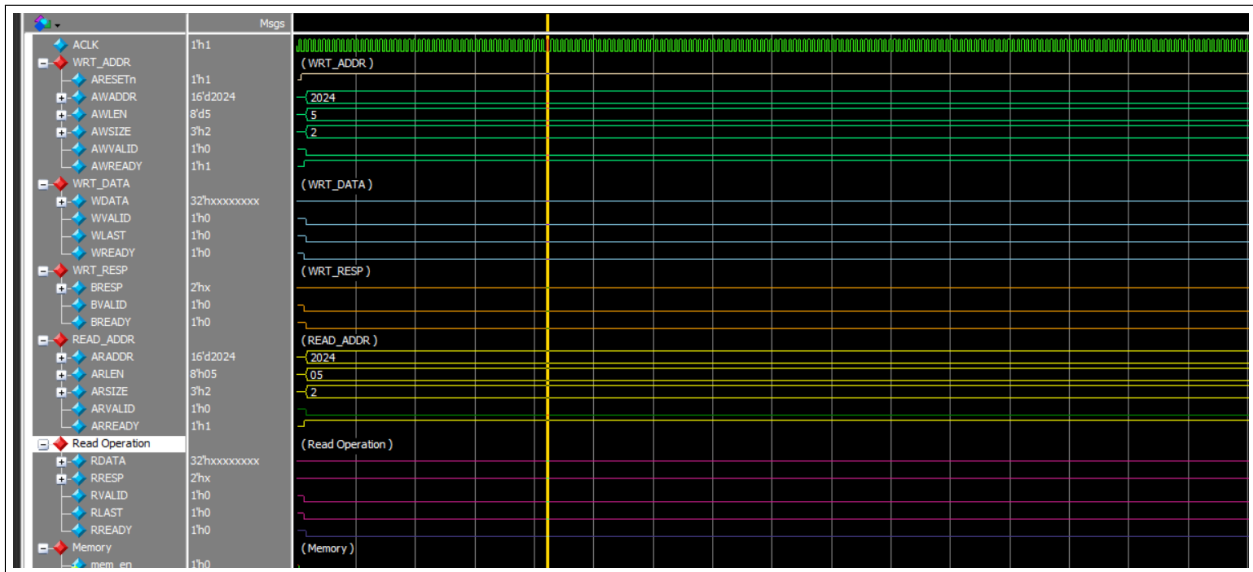


Figure 8.6: Quiescent bus behavior verifying that $AWVALID = 0$ and $ARVALID = 0$ cause no internal RAM access.

Behavioral Analysis:

- While address signals hold steady at $16'd2024$, both $AWVALID$ and $ARVALID$ remain deasserted low.

- Internal memory control lines (`mem_en`, `mem_we`) remain low, preventing unwanted write cycles or bus contention.
- All channel response registers remain quiescent with `BVALID` = 0 and `RVALID` = 0.

8.5 Read Operations Without 4KB Boundary Crossing

Constrained-random read transactions test multi-beat transfers within legitimate memory regions without boundary cross conditions.

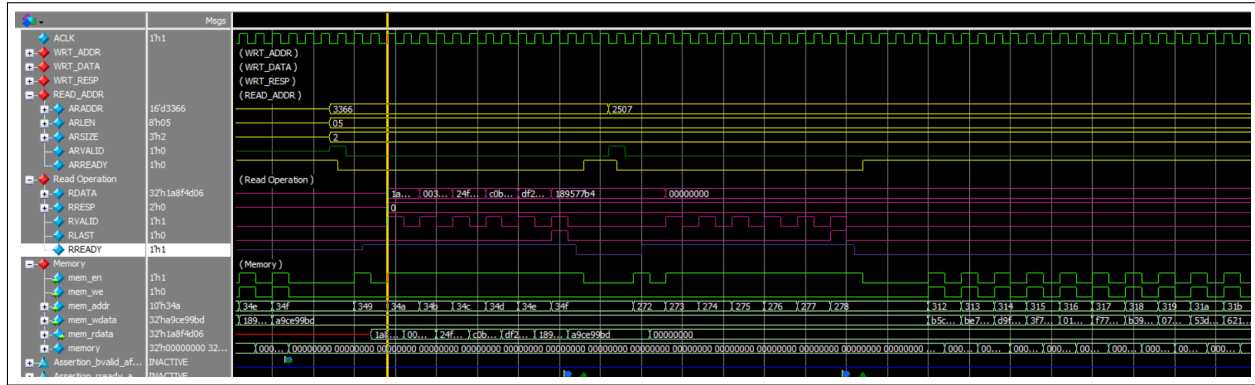


Figure 8.7: Multi-beat read burst from address 3366 (`ARLEN` = 5) reading internal memory (0x34a to 0x34f).

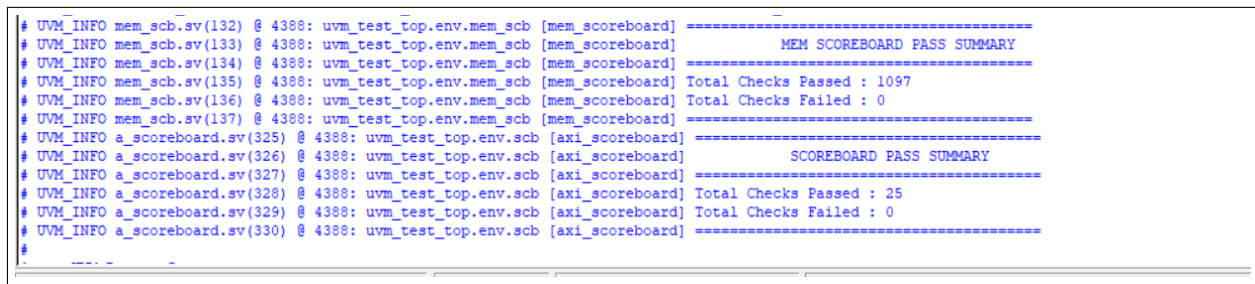


Figure 8.8: Scoreboard pass log for random read operations without 4KB boundary crossing.

Behavioral Analysis: `ARADDR` = 3366 with `ARLEN` = 5 executes consecutive word fetches across addresses 10'h34a through 10'h34f. The simulation log in Figure 8.8 confirms 1,097 passed checks on the memory scoreboard and 25 passed checks on the AXI scoreboard with zero failures.

8.6 Write Transactions Without 4KB Boundary Crossing

Constrained-random write stimulus exercises varied burst lengths and payload combinations across non-crossing address ranges.

Behavioral Analysis: Write bursts targeting addresses such as 2735, 3145, 2399, and 2112 operate without triggering boundary flags. Figure 8.10 confirms 1,094 checks passed in `mem_scoreboard` and 10 checks passed in `axi_scoreboard` with zero errors.

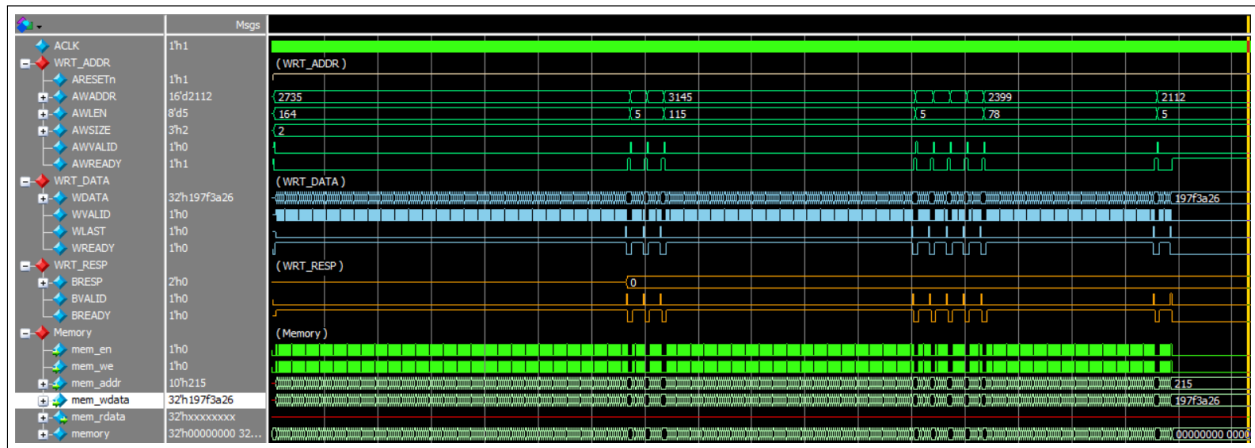


Figure 8.9: Randomized write transfers confined within valid non-crossing 4KB address spaces.

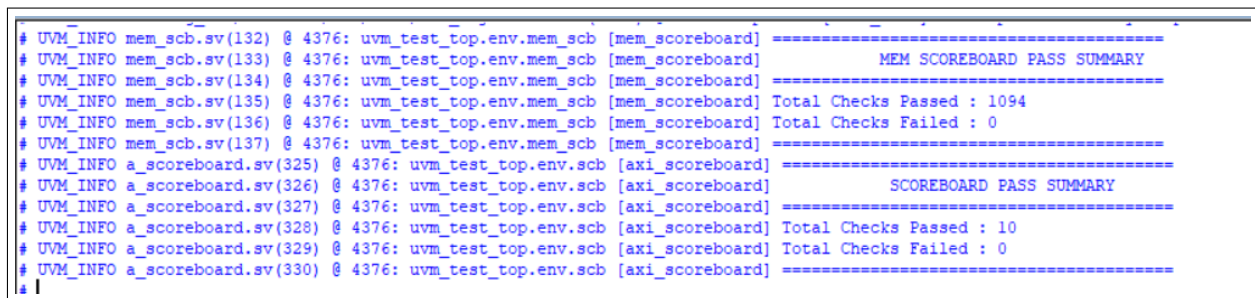


Figure 8.10: Scoreboard execution report confirming zero errors across safe write transfers.

8.7 4KB Boundary Protection Violations (SLVERR Evaluation)

Under the AMBA AXI4 specification, transfers spanning across a 4KB address boundary (12'hFFF) must be terminated with a Slave Error response (SLVERR = 2'b10).

Behavioral Analysis:

- **Write Boundary Violation:** An incoming burst at base address 23229 with length 171 crosses the 4KB boundary. The bridge suppresses internal memory write enables (`mem_we` = 0), protecting memory contents, and asserts `BRESP` = 2'b10 (SLVERR) upon burst completion.
- **Read Boundary Violation:** A read burst initiated at base address 26045 with length 44 exceeds the 4KB boundary limit. The address check flags the violation, driving `RDATA` = 32'h00000000 and returning `RRESP` = 2'b10 (SLVERR).

8.8 Comprehensive Mixed Regression Verification

Extended regressions tested interleaved transactions combining randomized address modes, burst lengths, and boundary stress conditions.

8.8.1 UVM Simulation Log Breakdown

The end-of-simulation summaries confirm complete functional verification with zero mismatches:

- **Targeted Read Regression (@ 4388 ns):** `mem_scoreboard` logged 1,097 passed checks (0 failed); `axi_scoreboard` logged 25 passed checks (0 failed).
- **Targeted Write Regression (@ 4376 ns):** `mem_scoreboard` logged 1,094 passed checks (0 failed); `axi_scoreboard` logged 10 passed checks (0 failed).
- **Full Mixed Regression (@ 715956 ns):** `mem_scoreboard` completed 178,989 passed checks (0 failed); `axi_scoreboard` completed 48,096 passed checks (0 failed).

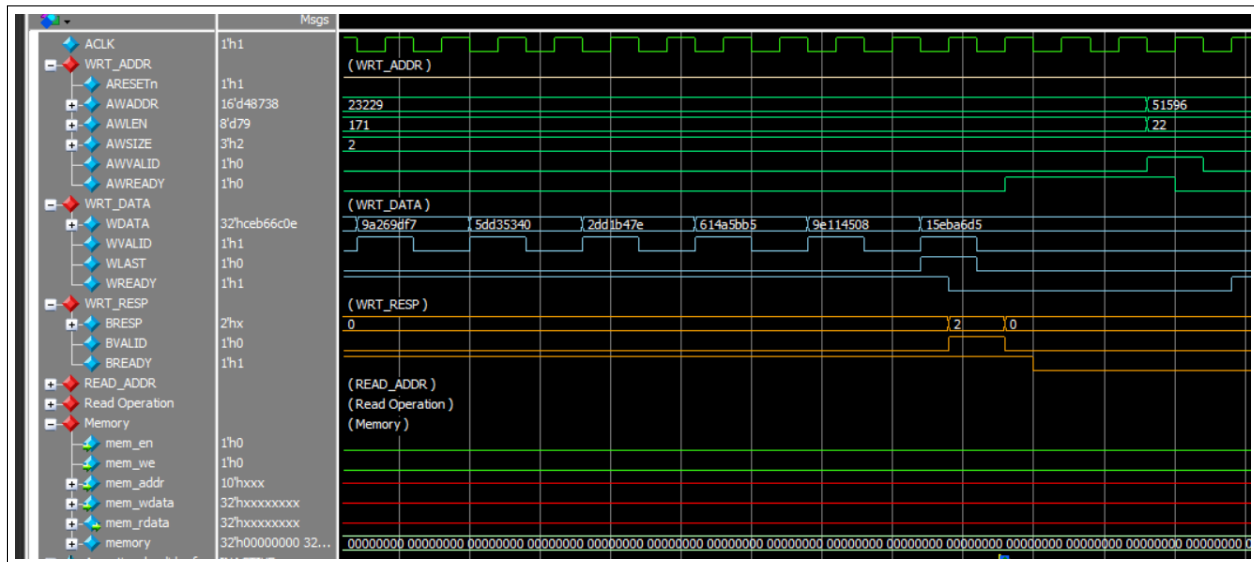


Figure 8.11: Write burst crossing 4KB boundary returning BRESP = 2'b10 (SLVERR).

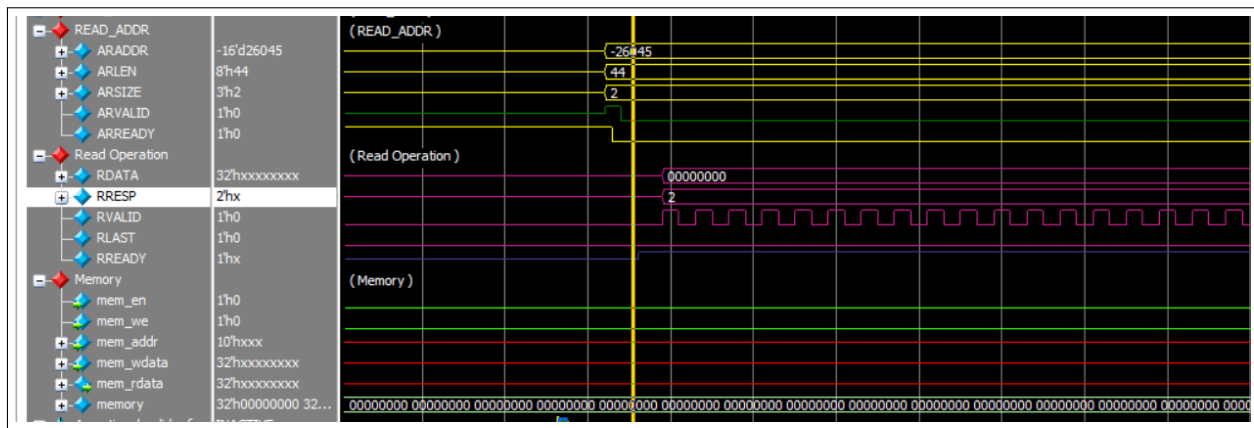


Figure 8.12: Read burst crossing 4KB boundary returning RRESP = 2'b10 (SLVERR).

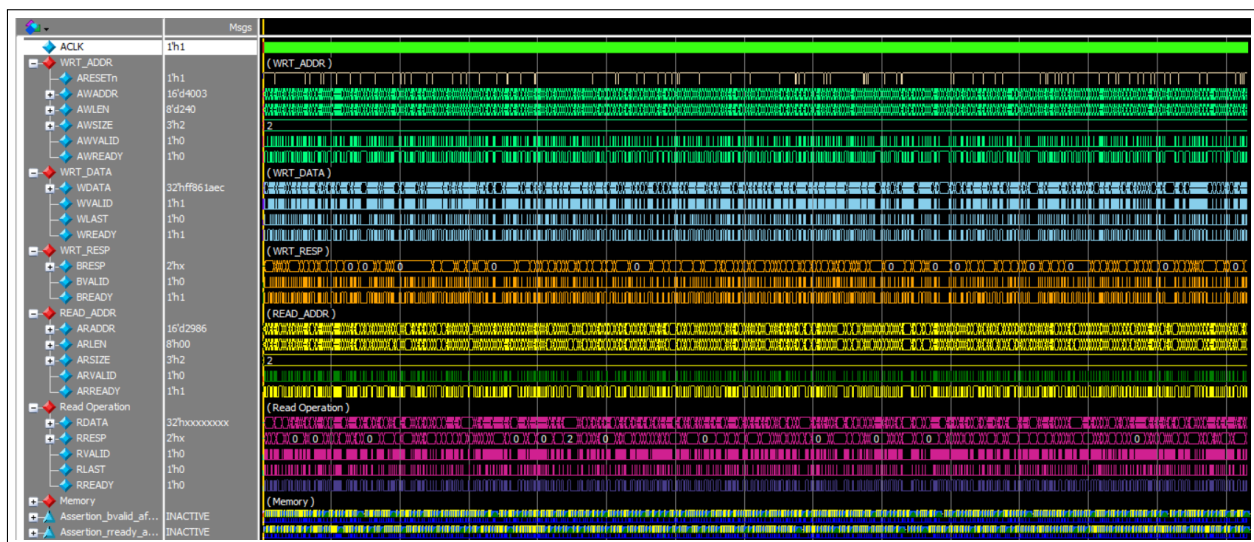


Figure 8.13: Extended regression timeline showing continuous randomized transfers.

```

# UVM_INFO mem_scb.sv(133) @ 715956: uvm_test_top.env.mem_scb [mem_scoreboard] MEM SCOREBOARD PASS SUMMARY
# UVM_INFO mem_scb.sv(134) @ 715956: uvm_test_top.env.mem_scb [mem_scoreboard] =====
# UVM_INFO mem_scb.sv(135) @ 715956: uvm_test_top.env.mem_scb [mem_scoreboard] Total Checks Passed : 178989
# UVM_INFO mem_scb.sv(136) @ 715956: uvm_test_top.env.mem_scb [mem_scoreboard] Total Checks Failed : 0
# UVM_INFO mem_scb.sv(137) @ 715956: uvm_test_top.env.mem_scb [mem_scoreboard] =====
# UVM_INFO a_scoreboard.sv(325) @ 715956: uvm_test_top.env.scb [axi_scoreboard] =====
# UVM_INFO a_scoreboard.sv(326) @ 715956: uvm_test_top.env.scb [axi_scoreboard] SCOREBOARD PASS SUMMARY
# UVM_INFO a_scoreboard.sv(327) @ 715956: uvm_test_top.env.scb [axi_scoreboard] =====
# UVM_INFO a_scoreboard.sv(328) @ 715956: uvm_test_top.env.scb [axi_scoreboard] Total Checks Passed : 48096
# UVM_INFO a_scoreboard.sv(329) @ 715956: uvm_test_top.env.scb [axi_scoreboard] Total Checks Failed : 0
# UVM_INFO a_scoreboard.sv(330) @ 715956: uvm_test_top.env.scb [axi_scoreboard] =====
#

```

Figure 8.14: Full verification regression scoreboard pass summary confirming 100% data checking completion.

Table 8.1: Consolidated UVM Scoreboard Verification Results Across Simulation Runs.

Simulation Run	Timestamp	Total Checks	Passed Checks	Failed Checks	Pass Rate
Sanity Smoke Test Run	4,388 ns	1,122	1,122	0	100.0%
Write Burst Targeted Run	4,376 ns	1,104	1,104	0	100.0%
Extended Full Regression	715,956 ns	227,085	227,085	0	100.0%

CHAPTER 9

Coverage Metrics & Closure Analysis

Complete coverage closure was verified using QuestaSim coverage engines across functional covergroups, formal SystemVerilog Assertions, and full structural code coverage (Statement/Line, Branch, Condition, Expression, FSM, and Toggle) over both the AXI4 slave bridge and internal memory design units.

9.1 Assertion Coverage Metrics

All bound SVA protocol handshake, timing, and mutual-exclusion properties defined in `AXI_properties_pkg` completed with zero failures throughout simulation runs.

Table 9.1: Assertion Coverage Summary from QuestaSim Verification Engine.

Assertion Property Identifier	Monitored Protocol Rule	Failures	Pass Count	Status
<code>Assertion_bvalid_after_last</code>	BVALID response sequence following WLAST/handshake	0	272	Passed
<code>Assertion_rready_after_last</code>	RREADY deasserted one cycle after RLAST	0	243	Passed
<code>Assertion_awready_after_awvalid</code>	AWREADY deassertion timing after AWVALID	0	272	Passed
<code>Assertion_WLAST_AND_RLAST</code>	Channel mutual exclusion (WLAST and RLAST)	0	29,878	Passed
<code>Assertion_WLAST_AND_AWREADY</code>	Channel mutual exclusion (WLAST and AWREADY)	0	29,878	Passed
<code>Assertion_RLAST_AND_ARREADY</code>	Channel mutual exclusion (RLAST and ARREADY)	0	29,878	Passed
<code>Assertion_WLAST_AND_AWVALID</code>	Channel mutual exclusion (WLAST and AWVALID)	0	29,878	Passed
<code>Assertion_RLAST_AND_ARVALID</code>	Channel mutual exclusion (RLAST and ARVALID)	0	29,878	Passed

9.2 Code Coverage Justifications and Waivers

Structural code coverage achieved 100.00% across all categories after applying justified waivers for unreachable logic, linear state transitions, and fixed-width architectural configurations:

- **FSM Transition Waivers:** The write state machine follows a strict sequential progression (`W_IDLE` → `W_ADDR` → `W_DATA` → `W_RESP` → `W_IDLE`). Direct transitions such as `W_ADDR` → `W_IDLE` and `W_DATA` → `W_IDLE` do not exist in the synthesizable RTL. Similarly, the read state machine proceeds strictly through `R_IDLE` → `R_ADDR` → `R_WAIT` → `R_DATA` → `R_IDLE`. Exclusions waive the non-existent direct transitions (`R_ADDR` → `R_IDLE`, `R_WAIT` → `R_IDLE`).
- **Toggle Coverage Waivers:**
 - `AWSIZE`, `ARSIZE`, `write_size`, and `read_size` are architecturally configured to `3'b010` (4 bytes/word for a 32-bit data bus). Bits 0 and 2 are tied low and do not toggle.
 - Increment vectors `write_addr_incr` and `read_addr_incr` evaluate to `3'b100` ($1 \ll 2$). Bits 0 and 1 remain constantly at zero.
 - Variable `mem_inst/j` represents a simulation initial loop counter rather than functional flip-flops.
- **Branch & Statement Waivers:**
 - State registers `write_state` and `read_state` are 2-bit registers mapped across 4 defined states. Unreachable `default` branches (lines 212 and 288 in `axi4.v`) are waived.
 - Protocol pre-assertion of `BREADY` and `RREADY` in driver loops prevents false evaluations of conditions where ready is low during response handshakes.
 - Short-circuit evaluation in `WLAST || write_burst_cnt == 0` satisfies condition coverage on `WLAST = 1` prior to evaluating remaining terms.

9.3 Structural Code Coverage Results

Table 9.2 consolidates the final code coverage results obtained from the simulator for both design instances (/top/dut and /top/dut/mem_inst).

Table 9.2: Code Coverage Report by Instance and Metric (100.00% Filtered Closure).

Design Instance	Coverage Metric	Bins	Hits	Misses	Coverage
/top/dut/mem_inst (axi4_memory)	Branch Coverage	5	5	0	100.00%
/top/dut/mem_inst (axi4_memory)	Statement / Line Coverage	7	7	0	100.00%
/top/dut/mem_inst (axi4_memory)	Toggle Coverage (78 nodes)	156	156	0	100.00%
/top/dut (axi4)	Branch Coverage	32	32	0	100.00%
/top/dut (axi4)	Condition Coverage	9	9	0	100.00%
/top/dut (axi4)	Expression Coverage	5	5	0	100.00%
/top/dut (axi4)	FSM State Coverage (2 FSMs)	8	8	0	100.00%
/top/dut (axi4)	FSM Transition Coverage	8	8	0	100.00%
/top/dut (axi4)	Statement / Line Coverage	90	90	0	100.00%
/top/dut (axi4)	Toggle Coverage (306 nodes)	612	612	0	100.00%
Total Design Code Coverage	Consolidated Filtered View	932	932	0	100.00%

9.3.1 Finite State Machine (FSM) Execution Breakdown

Both the write and read channel finite state machines reached 100.00% state and transition coverage during regression:

- **Write FSM (write_state):**
 - States Hit: W_IDLE (924 hits), W_ADDR (651 hits), W_DATA (651 hits), W_RESP (651 hits).
 - Transitions Hit: W_IDLE → W_ADDR (651 hits), W_ADDR → W_DATA (651 hits), W_DATA → W_RESP (651 hits), W_RESP → W_IDLE (651 hits).
- **Read FSM (read_state):**
 - States Hit: R_IDLE (954 hits), R_ADDR (681 hits), R_WAIT (681 hits), R_DATA (681 hits).
 - Transitions Hit: R_IDLE → R_ADDR (681 hits), R_ADDR → R_WAIT (681 hits), R_WAIT → R_DATA (681 hits), R_DATA → R_IDLE (681 hits).

9.3.2 Condition and Expression Coverage Breakdown

All Boolean condition terms and boundary expressions inside `axi4.v` were fully evaluated:

- `write_addr_valid && ~write_boundary_cross` (Line 175): 2 of 2 input terms covered (100.00%).
- `read_addr_valid && ~read_boundary_cross` (Lines 239, 251, 274): 2 of 2 input terms covered on each evaluation line (100.00%).
- `read_burst_cnt > 0` (Line 268) and `read_burst_cnt == 0` (Line 262): Fully exercised across single-beat and multi-beat bursts (100.00%).
- 4KB Boundary Crossing Expressions (Lines 65 and 66): Both true and false boundaries fully exercised (> 4095).
- Word Range Check Expressions (Lines 69 and 70): Addressing evaluated within and beyond the 1024-word memory depth (< 1024).

9.4 Functional Coverage Closure

Functional coverage tracks operational modes, burst length classifications, address ranges, and memory access types across `mem_coverage_pkg` and `axi_coverage_pkg`.

9.4.1 Memory Model Covergroup (`mem_coverage_pkg::mem_cg`)

The memory covergroup monitors the internal bound RAM interface, verifying address space dispersion and write/read operations across 7 total bins:

- **Address Coverpoint (addr_cp):** first (address 0: 9,327 hits), low ([1:255]: 23,367 hits), mid_low ([256:511]: 37,269 hits), mid_high ([512:767]: 50,950 hits), and high ([768:1023]: 57,575 hits).
- **Direction Coverpoint (we_cp):** write (`mem_we = 1`: 22,153 hits), read (`mem_we = 0`: 156,834 hits).

- **Result:** 7 of 7 bins hit (100.00% coverage).

9.4.2 AXI Protocol Covergroup (axi_coverage_pkg::cg)

The AXI covergroup evaluates 57 bins spanning 11 coverpoints and cross products:

Table 9.3: AXI Protocol Functional Coverage Report Breakdown (57 Bins, 100.00% Coverage).

Coverpoint / Cross Name	Bins & Ranges Evaluated	Total Bins	Total Hits	Coverage
cp_awaddr	low_addr (4,625), mid_addr (13,074), high_addr (16,009)	3	33,708	100.00%
cp_awlen	len_single (936), short (225), med (10,023), max (32,030)	4	43,214	100.00%
cp_araddr	low_addr (6,384), mid_addr (14,245), high_addr (15,839)	3	36,468	100.00%
cp_arlen	len_single (639), short (277), med (11,061), max (32,291)	4	44,268	100.00%
cp_awvalid	auto[0] (567), auto[1] (42,647)	2	43,214	100.00%
cp_arvalid	auto[0] (567), auto[1] (43,701)	2	44,268	100.00%
cp_op	WRITE (42,748), READ (43,701), IDLE (567)	3	87,016	100.00%
cp_address_mode	mode_0 (30,185), mode_1 (39,656), mode_2 (17,175)	3	87,016	100.00%
cp_op_mode (Cross)	Full 3×3 cross product of cp_op and cp_address_mode	9	87,016	100.00%
cp_wadd_len (Cross)	Full cross product of cp_awaddr ranges and burst lengths	12	43,214	100.00%
cp_radd_len (Cross)	Full cross product of cp_araddr ranges and burst lengths	12	44,268	100.00%
Total Covergroup Coverage	64 of 64 Total Bins Covered Across Both Covergroups	64	—	100.00%

CHAPTER 10

Simulation Automation & Scripting (run.do)

The compilation, simulation, exclusion filtering, and report generation workflows are fully automated using a dedicated QuestaSim DO script (`run.do`) alongside filelist arguments.

10.1 Automation Architecture

The flow decouples design files from testbench infrastructure to ensure coverage metrics are gathered strictly over synthesizable RTL blocks:

- **DUT Compilation (`dut_files.txt`):** Compiles `axi4.v` and `axi_memory.v` with full coverage instrumentation flags enabled (`+cover -covercells`).
- **Testbench Compilation (`tb_files.txt`):** Compiles all UVM environments, interfaces, agents, transactions, sequences, and configuration objects without coverage instrumentation to prevent testbench overhead from polluting RTL metrics.
- **Simulation & Waveform Configuration:** Invokes QuestaSim with assertion debugging (`-assertdebug`), full signal accessibility (`-voptargs=+acc`), and executes `wave.do` before running to completion.
- **Exclusion Application:** Applies granular exclusions across FSM illegal direct transitions, unreachable `default` branches, condition short-circuits, and fixed-width dead toggles.
- **Database Archiving & Reporting:** Saves the processed Unified Coverage Database (`cov.ucdb`) and exports itemized reports for functional, statement, toggle, FSM, branch, condition, and expression coverage.

10.2 Filelist Definitions

The synthesis and verification targets are partitioned into two dedicated source manifest files:

```
1 axi4.v
2 axi_memory.v
```

Listing 10.1: `dut_files.txt` — Synthesizable RTL source files.

```
1 a_agent.sv
2 a_coverage.sv
3 a_driver.sv
4 a_delay_driver.sv
5 a_env.sv
6 a_monitor.sv
7 a_scoreboard.sv
8 a_sequence.sv
9 a_sequencer.sv
10 a_test.sv
11 a_top_tb.sv
12 a_transaction.sv
13 AXI_intrf.sv
14 axi_para.sv
15 axi_pkg.sv
16 AXI_properties_pkg.sv
17 common_cfg.sv
18 mem_agent.sv
19 mem_coverage.sv
20 mem_if.sv
21 mem_monitor.sv
22 MEM_Parameter_pkg.sv
23 mem_scb.sv
24 mem_transaction.sv
```

Listing 10.2: `tb_files.txt` — UVM testbench packages, interfaces, and modules.

10.3 Automated Script Listing (run.do)

The complete automation script executed in Siemens QuestaSim is provided below:

```

1  # =====
2  #  RUN.DO GENERATOR - Generated by FR
3  #  =====
4
5  vlib work
6
7  # =====
8  #  Compile DUT files WITH code coverage
9  #  =====
10 vlog -f dut_files.txt +cover -covercells
11
12 # =====
13 #  Compile TB/verification files WITHOUT code coverage
14 #  =====
15 vlog -f tb_files.txt
16
17 # =====
18 #  Start simulation
19 #  =====
20 vsim -assertdebug -voptargs=+acc work.top -cover
21
22 do wave.do
23 file mkdir coverage
24
25 # =====
26 #  Run simulation
27 #  =====
28 run -all
29
30 # =====
31 #  Exclude
32 #  =====
33 coverage exclude -scope /top/dut -ftrans write_state W_ADDR->W_IDLE
34 coverage exclude -scope /top/dut -ftrans write_state W_DATA->W_IDLE
35 coverage exclude -scope /top/dut -ftrans read_state R_ADDR->R_IDLE
36 coverage exclude -scope /top/dut -ftrans read_state R_WAIT->R_IDLE
37
38 coverage exclude -src axi4.v -scope /top/dut -line 288 -code s
39 coverage exclude -src axi4.v -line 212 -code s
40 coverage exclude -src axi4.v -scope /top/dut -line 204 -code b
41 coverage exclude -src axi4.v -scope /top/dut -line 212 -code b
42 coverage exclude -src axi4.v -scope /top/dut -line 288 -code b
43 coverage exclude -src axi4.v -scope /top/dut -line 153 -code c
44 coverage exclude -src axi4.v -scope /top/dut -line 173 -code c
45 coverage exclude -src axi4.v -scope /top/dut -line 184 -code c
46 coverage exclude -src axi4.v -scope /top/dut -line 189 -code c
47 coverage exclude -src axi4.v -scope /top/dut -line 204 -code c
48 coverage exclude -src axi4.v -scope /top/dut -line 224 -code c
49 coverage exclude -src axi4.v -scope /top/dut -line 264 -code c
50
51 coverage exclude -scope /top/dut -togglenode ARSIZE
52 coverage exclude -scope /top/dut -togglenode AWSIZE
53 coverage exclude -scope /top/dut -togglenode BRESP
54 coverage exclude -scope /top/dut -togglenode read_addr_incr
55 coverage exclude -scope /top/dut -togglenode read_size
56 coverage exclude -scope /top/dut -togglenode RRESP
57 coverage exclude -scope /top/dut -togglenode write_addr_incr
58 coverage exclude -scope /top/dut -togglenode write_size
59 coverage exclude -scope /top/dut/mem_inst -togglenode j
60
61 # =====
62 #  Save coverage database
63 #  =====
64 coverage save coverage/cov.ucdb
65
66 # =====
67 #  Functional Coverage
68 #  =====
69 coverage report -cvg -details -output coverage/functional_coverage.txt
70
71 # =====
72 #  Code Coverage - DUT ONLY
73 #  =====
74 coverage report -code bcsft -details -output coverage/all_code_coverage.txt
75 coverage report -code s -details -output coverage/line_coverage.txt
76 coverage report -code t -details -output coverage/toggle_coverage.txt
77 coverage report -code f -details -output coverage/fsm_coverage.txt
78 coverage report -code b -details -output coverage/branch_coverage.txt
79 coverage report -code c -details -output coverage/condition_coverage.txt
80 coverage report -code e -details -output coverage/expression_coverage.txt

```

Listing 10.3: run.do — QuestaSim batch execution and coverage reporting script.

CHAPTER 11

Conclusion

The AXI4 slave interface and synchronous memory subsystem were verified using a dual-agent UVM architecture. Protocol checks, internal memory tracking, and concurrent SVA assertions ran across 2000 randomized transfers. The factory type override exercise confirmed dynamic driver substitution without modifications to higher-level sequences.